

Deep Learning Approaches to Grasp Synthesis: A Review

Rhys Newbury[✉], Morris Gu, Lachlan Chumbley, Arsalan Mousavian[✉], Clemens Eppner[✉],
Jürgen Leitner[✉], *Member, IEEE*, Jeannette Bohg[✉], *Member, IEEE*, Antonio Morales[✉],
Tamim Asfour[✉], *Senior Member, IEEE*, Danica Kragic[✉], Dieter Fox, *Fellow, IEEE*, and Akansel Cosgun[✉]

Abstract—Grasping is the process of picking up an object by applying forces and torques at a set of contacts. Recent advances in deep learning methods have allowed rapid progress in robotic object grasping. In this systematic review, we surveyed the publications over the last decade, with a particular interest in grasping an object using all six degrees of freedom of the end-effector pose. Our review found four common methodologies for robotic grasping: sampling-based approaches, direct regression, reinforcement learning, and exemplar approaches. In addition, we found two “supporting methods” around grasping that use deep learning to support the grasping process, shape approximation, and affordances. We have distilled the publications found in this systematic review (85 papers) into ten key takeaways we consider crucial for future robotic grasping and manipulation research.

Index Terms—Dexterous manipulation, deep learning in robotics and automation, grasping, perception for grasping and manipulation.



Fig. 1. Grasp synthesis is the problem of creating a grasp pose or a set of grasp poses. Robotic manipulators can grasp objects from multiple angles. This systematic survey reviews deep learning approaches to grasp synthesis that generate grasps utilizing all six DoFs.

I. INTRODUCTION

THE manipulation of objects is an essential skill for a robot in the real world, with grasping being an integral part of such tasks. Grasping is the process of controlling an object’s motion in a desired way by applying forces and torques at a set of contacts. It is an essential ability required for the majority of object manipulation tasks. Synthesizing a good grasp proposal for a specific gripper and an object or a scene made of objects is a high-dimensional search or optimization problem as there are many relative gripper–object poses, joint configurations, and contact conditions. An example of multiple grasp proposals for a single object can be seen in Fig. 1. The quality of each of these *grasp hypotheses* can be evaluated under a variety of criteria such as grasp stability, which depends on factors such as object or scene geometry, gripper geometry, and kinematics, as well as suitability for a specific manipulation task. Reflecting on more than four decades of research in robotic grasping, we see a change in how grasping is formulated and studied.

Early work on robotic grasping developed a theoretical framework that forms the basis of analytical approaches to grasping [1]. At the core of this framework are contact models, which are typically based on point contacts that define what components of contact forces and torques (i.e., wrenches) can be transmitted at a specific contact and act on the object. In this framework, a *grasp* is defined as the set of wrenches that can be achieved on an object. The goal of grasp synthesis is then often framed [2] as finding a grasp that keeps the object in equilibrium

Manuscript received 6 March 2023; accepted 29 April 2023. Date of publication 13 June 2023; date of current version 4 October 2023. This paper was recommended for publication by Associate Editor J. Kelly and Editor M. Yim upon evaluation of the reviewers’ comments. (*Corresponding author: Rhys Newbury.*)

Rhys Newbury is with Monash University, Clayton, VIC 3800, Australia, and also with the Australian National University, Canberra, ACT 2601, Australia (e-mail: newbury777@gmail.com).

Morris Gu and Lachlan Chumbley are with Monash University, Clayton, VIC 3800, Australia (e-mail: morris.gu1@monash.edu; lachlan.chumbley@monash.edu).

Arsalan Mousavian and Clemens Eppner are with the NVIDIA Corporation, Seattle, WA 98105 USA (e-mail: amousavian@nvidia.com; ceppner@nvidia.com).

Jürgen Leitner is with the Monash University, Clayton, VIC 3800, Australia, and also with LYRO Robotics Pty Ltd., Clayton, VIC 3168, Australia (e-mail: juxi.leitner@monash.edu).

Jeannette Bohg is with the Stanford University, Stanford, CA 94305 USA (e-mail: bohj@stanford.edu).

Antonio Morales is with the Jaume I University, 12071 Castellón de la Plana, Spain (e-mail: morales@icc.uji.es).

Tamim Asfour is with the Karlsruhe Institute of Technology, 76131 Karlsruhe, Germany (e-mail: asfour@kit.edu).

Danica Kragic is with the KTH Royal Institute of Technology, 10044 Stockholm, Sweden (e-mail: dani@kth.se).

Dieter Fox is with the NVIDIA Corporation, Seattle, WA 98105 USA, and also with the University of Washington, Seattle, WA 98195 USA (e-mail: fox@cs.washington.edu).

Akansel Cosgun is with the Deakin University, Burwood, VIC 3125, Australia (e-mail: akansel.cosgun@gmail.com).

An online version of the survey is available at <https://rhys-newbury.github.io/projects/6dof/>

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TRO.2023.3280597>.

Digital Object Identifier 10.1109/TRO.2023.3280597

in the presence of disturbances (i.e., *fixturing*) or moves it in a specific way (i.e., *dexterous manipulation*). Bicchi and Kumar [2] also mention enveloping grasps that wrap the fingers and the palm around the object, achieving more restraining grasps (i.e., *power grasps*). However, the limitation of these analytical approaches is that they assume full knowledge of object shape and geometry, material properties, and dynamics parameters. However, in reality, this information is rarely directly observable but can only be inferred from partial noisy sensory data.

The increased application of data-driven approaches to computer vision has successfully transferred to robotic grasping [3], primarily addressing the complexity and uncertainty in visual perception. To better fit these computer vision techniques, the focus of robotic grasping shifted from concepts around multifingered contact-based representation to pose-based ones. Commonly abstracting the robot's end-effector as an ideal two-fingered gripper approaching the object “top-down.” That way, a grasp is parameterized by the position and orientation of the coordinate frame attached to the gripper or the robot wrist. Before that, the degrees of freedom (DoFs) of a grasp were attributed to the robot hand, its kinematic structure, and the ability to control finger movements. This simplification of grasp parameterization is further supported by the increased availability of robust and simple end-effectors—e.g., parallel jaw grippers and suction cups [4]—and underactuated or soft hands [5], [6] that simplify the control, compared to a dexterous hand.

For the purpose of this review, we reuse terms such as four- and six-DoF grasps, where the former encodes position in 3-D and single rotation of a coordinate system attached to the gripper, relevant for defining “top-down” grasps. Six-DoF grasps relate to approaches that consider the full pose of that coordinate system. Fig. 2 shows examples of both four- and six-DoF grasps, for both parallel jaw grippers and dexterous hands. Six-DoF grasp synthesis is particularly useful in applications that require precise and dexterous manipulation of objects, such as for pick-and-place tasks. However, it can also be more challenging than four-DoF grasp synthesis due to the increased complexity of the grasping problem.

The dominant framework in the field of computer vision is applying deep learning to images. The robotics community often benefits from advancements in computer vision as the robots are often equipped with RGB and/or depth cameras. Deep learning, particularly when combined with large-scale data, has enabled new robotic capabilities that traditional machine learning (ML) or analytical methods cannot match. These include end-to-end manipulation learning [7], ambidextrous grasping [8], in-hand dexterous manipulation [9], pick-and-place in clutter [10], as well as more recent works such as task-agnostic generalization to new tasks [11] and translating natural language commands to robot actions [12].

Deep learning methods are also crucial for six-DoF grasp synthesis because they can offer increased accuracy, flexibility, reduced manual engineering, and robustness to variability compared to traditional ML or analytical methods. They can learn complex and flexible models that adapt to a wide range of grasping scenarios, without requiring manual feature engineering. In addition, deep learning methods can account for variations in

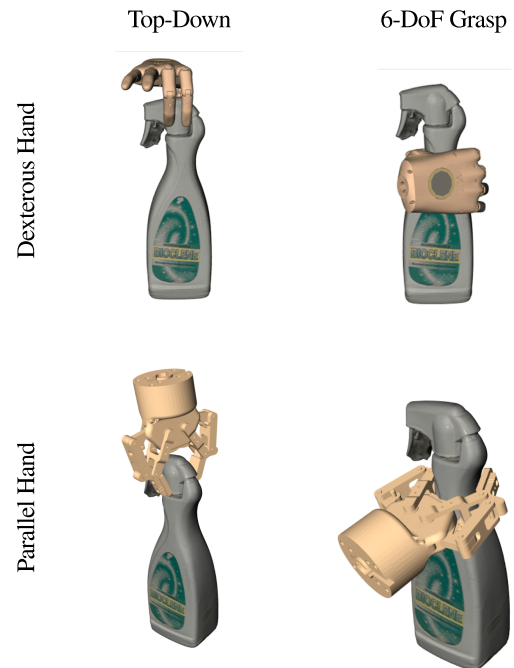


Fig. 2. This review focuses on the synthesis of six-DoF grasp pose hypotheses, where the DoF refers to the generated hand/wrist poses defined by the 3-D position and orientation of a gripper-specific coordinate system. This includes grasp synthesis with parallel grippers and dexterous hands alike.

object shape, pose, and other factors, resulting in more robust grasping performance. Typical stages for grasping an object are illustrated in Fig. 3, with grasp synthesis being the first stage before any action is taken. In this review, we focus on deep learning approaches applied to six-DoF grasping, in particular on the grasp synthesis stage of the grasping process.

There are recently published review papers [13], [14], [15] similar to ours. Kleeberger et al. [13] covers a broad range of methods for grasping, including model-based, analytical, optimization-based, and learning-based methods. While both papers cover similar topics, our article aims to provide a more detailed and comprehensive review on the state-of-the-art for six-DoF grasping. Kroemer et al. [14] provides a general review of ML techniques for robot manipulation, including grasping, reaching, and other manipulation tasks. The paper provides a high-level view of the field and is less focused on specific grasping methods. Differently from these two papers, ours has a specialized focus on the specific topic of deep learning for grasp synthesis while offering a comprehensive discussion around future research directions on the topic. To our knowledge, Duan et al. [15] is the closest to our work in terms of taxonomy, focusing on point cloud-based inputs and depth of discussion. The discussion in [15] is mainly around sampling-based methods, which was the dominant method before 2020, their cutoff year for the reviewed papers. In our article, in addition to sampling methods (see Section III-A), we capture the increasing popularity of end-to-end methods (called direct regression in our review—see Section III-B). We also identify supporting methods, such as affordances (see Section IV-B) and shape approximation (see Section IV-A), that are commonly

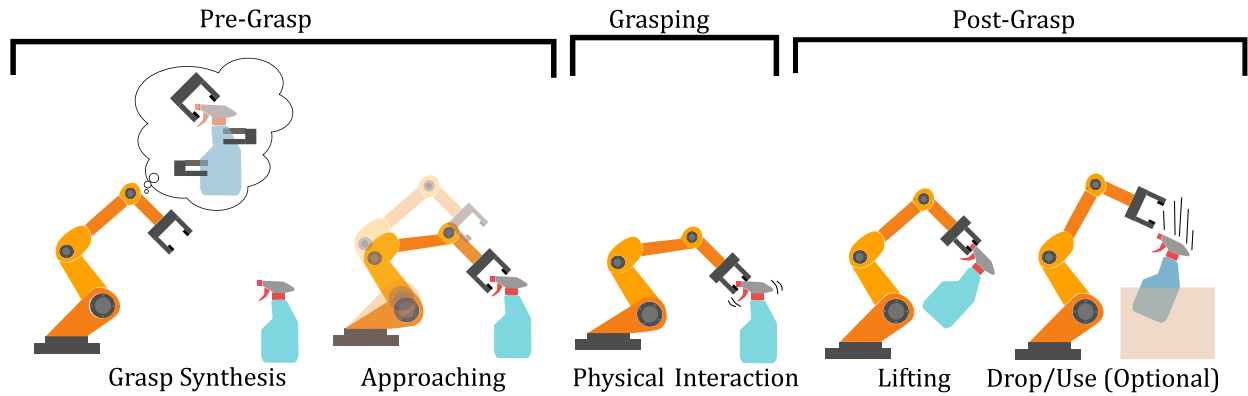


Fig. 3. Typical stages for grasping an object. Our review focuses on grasp synthesis, the first stage in the grasping process.

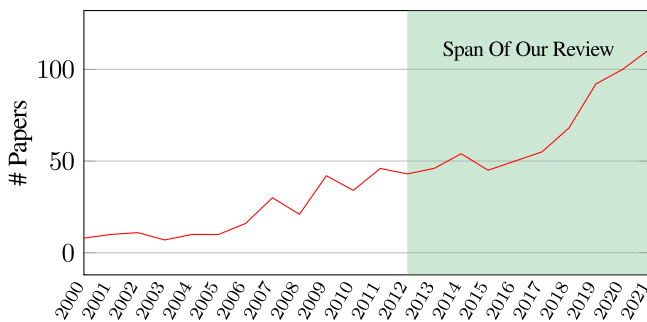


Fig. 4. Number of publications on IEEE Xplore that includes the keyword “Grasping” in metadata and “6DoF” in the full text is increasing year by year. We consider works published after January 1, 2012—when AlexNet [34] was published—in our review.

used together with main grasp synthesis methods. Differently from these review papers, ours provide an analysis on the benchmarking (see Section VI) and datasets (see Section V) for grasp synthesis. To our knowledge, our review is the only one that focuses on six-DoF grasp synthesis. We believe it is a crucial area of research for future progress in robotic manipulation.

Our systematic review is based on 85 research publications that employ deep learning methods for grasp synthesis, clustering the work along common methodologies, datasets, and object sets used. From the methods’ viewpoint, we devise a taxonomy along grasp sampling methods, direct regression methods, methods incorporating reinforcement learning (RL), and exemplar methods. Furthermore, we identify methods commonly employed in grasping, such as shape completion and affordances. Fig. 5 depicts a visual representation of the structure of the survey article.

The contributions of this survey are the following:

- 1) a systematic review of 85 papers, focusing on deep-learning-based six-DoF grasping;
- 2) the synthesis of the papers into ten key takeaways (discussed in Section VII), which we consider crucial for future research in robotics and manipulation.

II. NOTATIONS AND ANALYTICAL GRASPING

To provide a nuanced discussion regarding the contributions of deep-learning-based approaches to grasp synthesis, we briefly overview the necessary fundamentals of grasping and review notations and definitions used to define it.

- 1) A *grasp pose* defines the position and orientation of a grasp. Our survey found many different formulations of grasping, but most aim to learn the final pose of the robot to generate a successful grasp.
- 2) A *four-DoF grasp* defines a grasp where hand poses are generated and defined by a 3-D position and hand orientation about an approach vector that is commonly aligned with the direction of gravity and is, therefore, often referred to as “top-down grasping.” It is often denoted by x, y, z , and yaw .
- 3) A *six-DoF grasp* defines a grasp where hand poses are generated and defined by a 3-D position and orientation, thus six DoFs in total. The major difference to four-DoF grasps is a nonfixed approach vector, providing extra flexibility but increased complexity.
- 4) *Affordances* refers to the different tasks, which can be achieved with an object [16]. This definition has been adopted in previous grasping works (see, e.g., [17]).

An alternative way to frame a grasp, introduced by Morales et al. [18], is using the following three terms.

- 1) An *approach vector* defines the line along which the gripper or robotic hand approaches the target object.
- 2) The *grasp center point* is the point in space somewhere along the approach vector where the coordinate frame fixed to the gripper must be positioned before starting to close the fingers.
- 3) The *hand orientation* defines how the robot hand is oriented around the approach vector when placed on the grasp center point.

Some important terms are used throughout this article while describing approaches to grasping.

- 1) Each point force applied at a contact point on the object surface also generates a torque on the object. A *wrench* summarizes this pair of force and torque applied to the object through a contact in a 6-D vector.



Fig. 5. Visual representation of the core topics covered in this systematic survey.

- 2) A grasp is said to be in *force closure* if the forces that can be applied at the set of *frictional* contacts are sufficient to compensate for *any* external wrench applied to the object [19].
- 3) *Hand posture* describes the configuration of the gripper or hand fingers when the grasp is started or all the contacts are made
- 4) A *power grasp* is a grasp where there are multiple points of contact between the object and the fingers and palm. It maximizes the load-carrying ability of a grasp and is highly stable as the enveloping nature of the grasp provides force closure [20].
- 5) *Antipodal points* are pairs of points on the object surface whose normal vectors are collinear and pointing in the opposite direction [21]. With appropriate finger contact conditions, antipodal point grasps guarantee force closure.
- 6) An *antipodal grasp* is defined for two-fingered grippers that make contact with the object at antipodal points [21].

For consistency and a thorough discussion, we provide a short insight on the relevance of analytical approaches.

The majority of methods up to the year 2000 modeled grasping analytically [2], [3]. The focus was on modeling and estimating physical conditions of grasps, such as, for example, grasp stability. A force-closure grasp was often equated with a stable grasp, although force closure is a necessary but insufficient condition for a stable grasp [2], [22]. Physical conditions were

usually simplified through approximations such as point contact models, Coulomb friction, and rigid body dynamics [3], [22], [23]. Analytic approaches have been attributed to being complex and not applicable in real-time applications. However, analytic approaches address properties of grasps, while most of the recent methods for grasp synthesis focus on positioning the hand. The advantages of analytical methods are mathematical guarantees on grasp properties, such as force closure. This makes it easier to assess the conditions of grasps when objects are manipulated after a grasp has been applied. For example, what forces or torques can be exerted on the object before slippage occurs or how an object can be moved using in-hand manipulation. Within the first decade of the 21st century, there has been a rise of data-driven approaches to grasp synthesis [3], thanks to the development of grasping simulators such as GraspIt! [24]. Early approaches often used hand-designed features that corresponded to parts of objects that could be grasped [25], [26], [27]. Kamon et al. [28] presented one of the earliest works in four-DoF grasping that use ML for grasping objects. The authors hand designed a low-dimensional feature space to estimate the quality of grasps. Since then, many works have employed traditional learning methods with a larger feature space [29], [30], [31], [32], [33]. One key takeaway from this early research was that while grasping can be done using RGB data only [31], depth information (RGB-D) improved grasping success [29], [32]. We refer to [3] for a more in-depth review of earlier methods for

data-driven grasp synthesis. In this review, we focus on deep learning techniques in particular.

III. DEEP LEARNING METHODS IN SIX-DOF GRASPING

The number of publications investigating deep learning approaches for six-DoF grasping has grown significantly in the past few years, as highlighted in Fig. 4. From the systematic review, we identify four main algorithmic methodologies for grasp synthesis using deep learning based on the reviewed literature: grasp pose sampling, regressing grasp pose directly, RL, and exemplar methods. These procedures relate to how the grasps are generated at test time.

Sampling approaches consider one or many grasp samples and have learned a function to estimate the quality of a sampled grasp. An essential characteristic of sampling approaches is that each sample is evaluated *individually*. Alternatively, direct regression considers the data *globally* and learns a function to predict high-quality grasps. *RL* includes methods that involve the maximization of a cumulative reward function based on a robot's actions or demonstrations. Exemplar methods aim to use a similarity metric between grasps to retrieve high-quality grasps from an existing database, which are most similar. These categories are often mutually exclusive, though Aktaş et al. [35] employed both direct regression and sampling, while Mahler et al. [36] employed both sampling and exemplar methods.

A. Sampling

We define sampling methods as any approach that considers each sample *individually* and use information encoded about the sample to make decisions about the grasp. The samples may be sourced from any discrete or continuous n -dimensional space. All the reviewed sampling works are shown in Table I.

Deep learning approaches employing sampling implement the following steps: sample information; evaluate the sample according to a quality estimation function, parameterized by a deep neural network; and optionally refine the sample using an optimization-based approach to achieve a higher quality grasp. We adopt the term “quality”; however, there is no current consensus on the definition. Throughout this article, we use this term to refer to the confidence of grasp success. Table I presents an overview of all the *sampling* papers. A popular deep-learning-based approach for four-DOF grasping generates a series of antipodal grasps through sampling while using a neural network to predict grasp quality [69]. This aims to predict the probability of a successful grasp to generate a series of antipodal grasps.

1) *Generating Samples*: Authors can sample a subset of grasp parameters in an n -dimensional space. This is commonly done using one of two approaches. Random sampling occurs when samples are taken from an arbitrary random distribution. The most common distribution is a uniform distribution; however, some authors sample from other distributions such as Gaussian [50], [70]. The second method is where samples are taken at equispaced intervals within the space.

TABLE I
PUBLICATIONS WITH DEEP-LEARNING-FOCUSED SAMPLING METHODS

Year	Paper	Sample Space	Evaluation	Refinement
2015	Varley et al. [37]	Priors	Metric	
2015	Kappler et al. [38]	<i>Not Specified</i>	Binary	
2016	Gualtieri et al. [39]	Euclidian	Binary	
2017	ten Pas et al. [40]	Euclidian	Binary	
2017	Zhou and Hauser [41]	Euclidian	Binary	✓
2018	Yan et al. [42]	<i>Not Specified</i>	Binary	✓
2019	Mousavian et al. [43]	Latent	Binary	✓
2019	Liang et al. [44]	Priors	Metric	
2019	Lu et al. [45]	Euclidian	Binary	✓
2019	Ottenhaus et al. [46]	Priors	Metric	
2019	Gonçalves and Lima [47]	Euclidian	Binary	
2019	Aktaş et al. [35]	Priors	Binary	✓
2020	Riedlinger et al. [48]	Euclidian	Binary	
2020	Murali et al. [49]	Latent	Metric	✓
2020	Van der Merwe et al. [50]	Hand Posture	Binary	✓
2020	Lundell et al. [51]	Multiple Views	Binary	
2020	Lou et al. [52]	Euclidian	Metric	
2020	Choi et al. [53]	Multiple Views	Binary	
2020	Lu et al. [54]	Hand Posture	Binary	✓
2020	Schaub and Schöttl [55]	Multiple Views	Binary	
2020	Murali et al. [56]	Priors	Metric	
2020	Kokic et al. [57]	Euclidian	Metric	
2021	Lundell et al. [58]	Euclidian	Metric	
2021	Lou et al. [59]	Euclidian	Metric	
2021	Lundell et al. [60]	Euclidian	Metric	
2021	Peng et al. [61]	<i>Not Specified</i>	Metric	
2021	Jiang et al. [62]	Euclidian	Binary	
2021	Kasaei and Kasaei [63]	Multiple Views	Binary	
2021	Wang et al. [64]	Priors	Binary	
2021	Munoz [65]	Multiple Views	Binary	
2021	Corsaro et al. [66]	Priors	Binary	
2021	Ren et al. [67]	<i>Not Specified</i>	Metric	
2021	Wen et al. [68]	<i>Not Specified</i>	Metric	

The subset of grasp parameters can be sampled from Euclidean space, priors, configuration space, latent space, or multiple views. When sampling in Euclidean space, heuristic-based rules are often used to remove irrelevant grasp candidates.

a) *Euclidean space sampling*: One of the most common approaches is to sample a 3-D vector representing the translation part of the grasp pose. This is often achieved by sampling points from a point cloud [39], [52], [59]. The approach vector can be estimated by either using normal information estimated from the sensor data or sampling angles. The authors in [40] and [48] used the opposite direction of the surface normal to find the grasp approach vector. Riedlinger et al. [48] use a series of local augmentations on the initial grasp approach vector to generate a sample set of candidate approach vectors. Gualtieri et al. [39] generated candidate grasp approach vectors using the surface normal and curvature axis to generate equispaced orientations orthogonal to the curvature axis.

Some approaches sample angles independently of the surface normals. Lou et al. [52], [59] chose N points from a point cloud and sample the wrist angles randomly for the grasp. However, they restrict the approaching vector of the grasp to be above the table. Kokic et al. [57] randomly sample grasp and roll angles, and offset distances for each point in the point cloud. The authors in [45], [58], and [60] sample grasp candidates around the center of an object with a random orientation.

Sampling can also be performed with regularly spaced points through Euclidean space. Jiang et al. [62] sample regularly spaced points for the position of the grasp pose. Similarly, Gonçalves and Lima [47] sample equispaced points throughout the region of interest.

Alternatively, instead of sampling a xyz vector, angles can be used. Zhou and Hauser [41] chose a random hand orientation and approach vector, after which they translate the hand along the approach vector of each sample until a grasp is found that does not collide with the gripper when it is open but collides when closed.

In Euclidean space, sampled grasps are commonly pruned to remove infeasible grasps based on rules such as the following.

- 1) The robot hand is in collision with the point cloud when fingers are open [40], [41].
- 2) The closing region of the fingers does not contain at least one point from the point cloud [39], [40], [41].

b) Sampling priors: More complex algorithms can also be used to find a set of feasible grasp candidates. These are often used when creating grasp samples, which will then be evaluated for affordances [56] or different types of grasps (e.g., power and pinch grasps) [66]. The most commonly [44], [56], [64], [66] used grasping algorithm is a support-vector-machine-based approach proposed by Pas and Platt [71]. Mahler et al. [36] use a modification of the grasping algorithm proposed by Smith et al. [72] to generate a series of antipodal grasps. They frame the problem as a multiarmed bandit, to be solved using Thompson sampling [73].

While grasping simulators are often used for generating training data, they have also been used at test time to sample grasps. A common prerequisite for grasping simulators is a full approximation of the shape model. The common simulators for this purpose were GraspIt! [24] or Simox [74]. Ottenhaus et al. [46] generated grasp samples on a reconstructed object using the grasp planner by Simox [74]. Alternatively, Varley et al. [37] sample grasps using the simulated annealing planner [75] for partially visible objects.

A deep learning approach can also be used to synthesize grasp samples. Aktaş et al. [35] used the direct-regression-based approach from [76] to generate multiple grasps, to be used as grasp samples.

c) Latent space sampling: Mousavian et al. [43] train a variational autoencoder [77] and uniformly sample through latent space to generate grasp poses. An example of the grasps generated by this approach is shown in Fig. 6. This sampling approach is also adopted by Murali et al. [49].

d) Hand posture space sampling: Some authors sample grasp configurations from a prior distribution fit to the training set to create an initial hand configuration for their approach [50], [54], [70]. Van der Merwe et al. [50], [70] fitted a Gaussian mixture model to represent a grasp prior function trained on grasp configurations seen during training. Similarly, Lu et al. [54] trained a mixture density network [78] over all grasp attempts from the training set.

e) Multiple views: A number of papers employ multiple viewpoints of the scene, where these can be from either virtual or real cameras. A camera viewpoint is then sampled, and

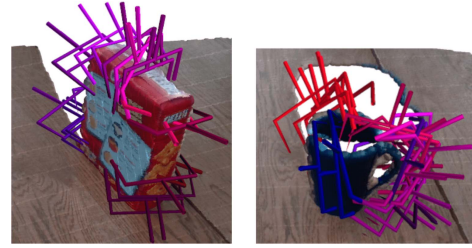


Fig. 6. Mousavian et al. [43] sample through the latent space of a trained model to generate a series of grasp candidates (©2019 IEEE).

a grasping approach is employed on the sampled viewpoint. Schaub and Schöttl [55] combine multiple viewpoints around the scene to create a 3-D representation of an object. Using the 3-D representation, they generate depth images for a series of virtual cameras. They then use a four-DoF grasping algorithm [79] for each real and virtual camera. This is extended by Schaub et al. [80] who fused depth images around the scene to provide more detailed depth maps. Choi et al. [53] extended a previous four-DoF approach [81] to six-DoF by using an iterative improvement algorithm approach to choose an approach direction. The authors in [63] and [65] generate multiple views of the object from virtual cameras using a captured point cloud from a single viewpoint. They proposed a method to select a view according to an entropy-based measure. Lundell et al. [51] used the algorithm proposed by Satish et al. [81] on the depth map from multiple viewpoints. The robot then executes the best grasp from all viewpoints.

2) Sample Evaluation: Once samples are generated, these approaches use a function, commonly parameterized by a neural network, to estimate a numerical value representing a grasp metric. This is commonly used an estimation of the “quality” of the grasp. The grasp quality can be approximated through an analytical metric or the likelihood of a grasp being successful. While some approaches execute the highest quality sample directly, others refine the samples using optimization-based techniques based on the learned quality function.

a) Binary classification: Predicting grasp success can be treated as a binary classification problem, where the output represents the confidence of a grasp being successful or not. A convolutional neural network (CNN) can be trained for the binary classification of grasp success, with the network outputting either 0 or 1 for a successful grasp [35], [38], [39], [40], [41], [42], [43], [45], [47], [48], [50], [54], [64], [66]. During the collection of ground truth training data to determine if a grasp is successful, the grasp is most commonly executed using a robotics simulation environment [35], [38], [41], [42], [43], [45], [48], [50], [54], [62], [66], where a grasp will be labeled successful if the object remains within the gripper after being lifted. Grasp success can also be based on an analytical metric, such as whether the grasp would form a force-closure grasp [40], [47] or if the grasp is antipodal [39].

The binary classification of a sampled grasps can also be considered as one of the outputs of a sampling approach. For example, Jiang et al. [62] designed an approach that learns implicit functions that predicts grasp parameters (quality, rotation,

and width) from the feature space representation of a randomly sampled query point. Van der Merwe et al. [50] train a network to predict the success of a grasp, as well as a signed distance function that represents a distance between a query point and the surface of the object.

b) Learning a metric: Learning metrics associated with grasps, rather than a pure binary label, have been proposed to better represent the quality of a grasp. These metrics are often continuous numbers (rather than the previously described binary label) that provide additional information to rank grasp quality or help “guide” deep learning algorithms using them as a fitness score.

Varley et al. [37] learned a series of heatmaps, where each pixel represents the location’s efficacy as a fingertip or palm location for common grasp types found in their training set. Liang et al. [44] designed an approach that learns a grasp quality metric based on the force-closure metric and wrench space analysis [82]. Similarly, Ottenhaus et al. [46] trained a CNN to estimate the force-closure probability of a grasp under a small random perturbation. Lundell et al. [58], [60] trained a grasp classifier using a *generative adversarial networks* and use the discriminator loss to help produce realistic-looking grasps. Wen et al. [68] compute a continuous score for each grasp by looking at the stability of randomly sampled grasps in the proximity of the selected grasp. They argue that “grasp stability should be continuous over its 6-D neighborhood”; therefore, this should allow for more stable grasps. Furthermore, this can allow for imperfections in the robotic grasp position while still being likely to execute a stable grasp. The authors in [61] and [67] design an approach to learn the smallest coefficient of friction, which will satisfy a force-closure grasp. Learned metrics can also represent the quality of the grasp with respect to other aspects of the task, for example, metrics based on the relevance of the grasp to a given affordance [56], [57].

Other approaches consider workspace constraints. Lou et al. [52] design an approach to learn the probability of grasp success and use an additional network to learn the probability of a grasp being reachable. These probabilities can be multiplied to find the likelihood of success for the entire grasping action. This work was extended by Lou et al. [59] to allow the robot to grasp in constrained environments, such as in boxes, where walls may limit the success of grasping an object.

3) Optimization-Based Grasp Refinement: Gradient-based optimization through the trained network can be used to find high-quality grasp candidates. A sampled grasp is taken as the initial seed, which is then refined based on the derivatives of the quality estimation network. This attempts to maximize estimated grasp quality. Zhou and Hauser [41] trained a CNN to predict grasp quality given a depth image and a sampled end-effector pose. The end-effector pose is locally optimized using the quasi-Newton method on the gradient of the learned quality function. Similarly, some authors take the derivative of the grasp quality with respect to the grasp pose and then use gradient ascent to refine the grasp candidates [42], [43], [45], [49]. An example of the refinement process is shown in Fig. 7, where the initial grasp is a potentially bad sample (blue) and is refined to a higher quality grasp (yellow). The authors in [50] and [54]



Fig. 7. Example of a grasp refinement process from [43], where the initial grasp is shown in dark blue and the final grasp pose is shown in yellow (©2019 IEEE).

treat finding a grasp configuration as an optimization problem aiming to maximize the probability of grasp success, seeding the process with samples from a prior distribution. Lu et al. [70] extend this using active learning during training, improving their results. In contrast to other works, Aktaş et al. [35] found that the optimization of the sample did not improve the grasp success rate in simulation. The authors explore the use of gradient-based optimization and simulated annealing for their optimization.

B. Direct Regression

Direct regression approaches can simultaneously process the entire sample space using a single network to predict either a single or multiple grasps, along with specific grasp properties such as parameters and quality, from visual information.

Direct regression approaches, which can be considered as end-to-end methods, utilize a single network to process the entire input to regress an output. These methods attempt to reduce computational cost compared to sampling methods by processing data globally through the network in a single pass. Early works in direct regression based on deep learning are inspired by object detection work from the computer vision community [106], [107]. Authors would treat finding top-down grasps similar to detecting objects in an image and would use the depth image to recover the 3-D position of the grasp [108], [109], [110], [111]. Morrison et al. [79] designed a heatmap-based approach for four-DoF grasping that was designed for fast inference, enabling closed-loop grasping and grasping of moving objects. However, as the dimensionality of the output increased, the problem difficulty also increased [43], [112]. This presents scaling issues when directly regressing to high-dimensional outputs such as a six-DoF grasp pose or a high-DoF dexterous robotic hand configuration. To overcome this, researchers often reduce the DoF of the output from a single network. This can be accomplished through multistage approaches, where each stage has a specific task, or by reducing the DoF by conditioning the output on an input. The direct regression approaches found in our systematic review are shown in Table II.

a) Directly regressing a pose: Schmidt et al. [84] present one of the earliest CNN-based works to directly estimate a single six-DoF grasp pose of an end-effector from input depth images. Similarly, Yang et al. [88] trained a network to estimate the transformation matrix needed to be applied to the end-effector to produce a successful grasp. However, both the approaches

TABLE II
APPROACHES TO DIRECT REGRESSION OF SIX-DOF GRASPING USING DEEP
LEARNING, GENERATING EITHER A SINGLE OR MULTIPLE GRASP POSES
AS AN OUTPUT

Year	Paper	Grasps	Direct Pose	Multistage	DimRed
2017	Veres et al. [83]	Single	✓		
2018	Schmidt et al. [84]	Single	✓		
2018	Choi et al. [85]	Single			✓
2019	Liu et al. [86]	Single	✓		
2020	Qin et al. [87]	Multiple			✓
2020	Yang et al. [88]	Single	✓		
2020	Jeng et al. [89]	Multiple			✓
2020	Wang and Lin [90]	Single			✓
2020	Fang et al. [91]	Multiple		✓	
2020	Wu et al. [92]	Multiple			✓
2020	Ni et al. [93]	Multiple			✓
2020	Breyer et al. [94]	Multiple			✓
2020	Liu et al. [95]	Single	✓		
2020	Shao et al. [96]	Single		✓	
2020	Ni et al. [97]	Multiple		✓	
2021	Wang et al. [98]	Multiple		✓	
2021	Sundermeyer et al. [99]	Multiple			✓
2021	Zhao et al. [100]	Multiple		✓	
2021	Gou et al. [101]	Multiple			✓
2021	Zhu et al. [102]	Multiple		✓	
2021	Wei et al. [103]	Multiple		✓	
2021	Li et al. [104]	Multiple			✓
2021	Li et al. [105]	Single		✓	

We found three main approaches: regressing to a pose directly (Pose), employing a multistage approach, or performing dimensionality reduction (DimRed).

assume that there is only a single most optimal ground truth grasp for each input. This would introduce ambiguities as there may potentially be numerous successful grasps that can be executed for any input. To account for this, Liu et al. [86] designed a loss function that accounts for multiple ground truth grasps in the training data by calculating the loss between the current output and the closest ground truth grasp. This is extended in another work [95] that includes differentiable terms for a grasp metric and self-collisions in the loss function. This allows their approach to work both in an unsupervised manner or using a smaller supervised dataset. Veres et al. [83] created a generative model based on a *conditional variational autoencoder* (CVAE) [113]. They use the CVAE to generate contact positions and contact normals for a multifingered robotic hand.

b) Reducing DoF: Due to the difficulty of regressing all six DoFs of a grasp, some DoFs can be reduced by analytically determining some DoFs conditioned on the regressed DoF. This is seen in some four-DoF approaches; for example, the depth of the grasp is not directly regressed but instead uses the depth image to recover the grasp depth conditioned on the grasp position and two-finger gripper wrist rotation [79].

Sundermeyer et al. [99] reduced the six-DoF grasping to a four-DoF representation by ensuring that one of the contact points for a two-finger parallel gripper is taken from the point cloud. The three-DoF rotation and gripper width is then estimated. Breyer et al. [94] directly output the predicted grasp quality, orientation, and opening width for each voxel in a queried 3-D volume. The 3-D position is recovered from the center of a voxel. Jeng et al. [89] proposed a coarse-to-fine

representation, where the orientation is initially coarsely discretized as a grid with a given confidence value. A refinement step is also used for each grasp pose to allow further flexibility from the discretized coarse representation. Gou et al. [101] estimate an $SO(3)$ orientation and confidence of every pixel directly from an RGB image. An analytical method based on the depth image is then used to find the gripper width and position for each pixel. Ni et al. [93] directly regressed grasps from sparse point clouds. They predict the grasp quality, approaching direction, and opening direction of the gripper for every point in the original point cloud. Likewise, the approach by Li et al. [104] estimates the rotation, width, depth, and quality of a grasp for each point in the point cloud. They combine this with another branch, which predicts whether the grasp would be in collision. Alternatively, Choi et al. [85] predict the most likely grasping direction and wrist orientation from a set of discretized directions and orientations. The translational part of the pose is found by determining the centered contacting voxel within the grasping direction. Wang and Lin [90] assume they are grasping the centroid of the object and attempt to directly regress the quaternion for the grasp. Wu et al. [92] create a network with three branches, trained to predict the likelihood of the grasp consisting of antipodal points, grasp offset for each of the sampled $SE(3)$ points, and grasp confidence (trained on whether grasp was executed in simulation). Qin et al. [87] trains a neural network based on [114] to predict both grasp point and quality for every point in a point cloud.

c) Multistage approach: These approaches use multiple stages, where a stage refers to a component of the end-to-end model with a loss function and at least one specific task. This aims to simplify learning by breaking the problem down into smaller parts. Most multistage approaches consider three-stage approaches [91], [98], [100], [102], [105], while Wei et al. [103] propose a two-stage approach. Stages are commonly done in series (one after another); however, the last two stages in the work by Fang et al. [91] are in parallel (simultaneously).

The first stages of the network are commonly used for at least one of the following: predicting grasp quality for subsampled points [91], [98], [100], estimating a grasp for each point [97], [103], estimating a subset of the DoF [91], or employing contrastive loss functions [102]. The middle stages can be used to create grasp proposals (if they were not generated in a previous stage) [98], [100] or further estimate a subset of the DoF [102].

The last stage can act as a refinement stage to improve the regressed grasps [97], [100], [103]. Fang et al. [91] used two final parallel stages to generate grasp proposals and predict the ability of the grasp pose to tolerate larger errors, aiming to improve the robustness to imperfect sensing or control. The final stage can also be used to predict the remaining DoFs [102]. Example grasps regressed in the work by Fang et al. [91] from real-world data are shown in Fig. 8.

An alternative multistage approach is demonstrated by the authors in [96] and [105], who propose an approach that uses each stage of the network to predict a single contact point. The subsequently regressed contact point will be conditioned on the previous point. The final grasp is then recovered from the

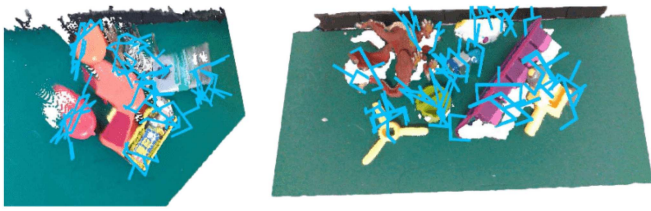


Fig. 8. Fang et al. [91] use a direct regression approach to generate a series of grasps from real-world data (©2020 IEEE).

TABLE III
OUR SYSTEMATIC SURVEY FOUND TEN PUBLICATIONS EMPLOYING *OFF*
POLICY OR *ON POLICY* RL

Year	Paper	Learning	Algorithm	LfD
2018	Gualtieri and Platt [120]	On-Policy	DQN[121]	
2019	Wu et al. [122]	On-Policy	PPO[123]	
2019	Merzic et al. [124]	On-Policy	TRPO [125]	
2020	Mandikal and Grauman [126]	On-Policy	PPO[123]	
2020	Song et al. [127]	Off-Policy	Q Learning[128]	✓
2020	Wu et al. [112]	On-Policy	PPO[123]	
2021	Berscheid et al. [129]	On-Policy	Single Step MDP	
2021	Kawakami et al. [130]	On-Policy	PPO[123]	✓
2021	Tang et al. [131]	Off-Policy	Q Learning[128]	
2021	Wang et al. [132]	Off-Policy	DDPG[133]	✓

The LfD column indicates approaches that learn from expert demonstration.

regressed contact points. Furthermore, Shao et al. [96] show that this approach is generalizable between different robotic hands.

C. Reinforcement Learning

RL approaches aim to learn a policy to maximize the cumulative reward commonly over a multistep task. We only review deep RL, where the policy is parameterized by a deep neural network. We subdivided the reviewed works into two main approaches: On- and Off-Policy learning. For a more comprehensive review on deep RL, see [115], and for a deeper exploration of RL in grasping and its open challenges, see [116].

Some seminal works in RL for grasping include the work in [117], [118], and [119]. Levine et al. [117] collected 800k grasp attempts over multiple months. They present a self-supervised approach, and the authors claimed that their approach is analogous to an RL formulation. The authors in [118] and [119] present works that formulate grasping as an RL problem. Kalashnikov et al. [118] focused on real-world data collection, collecting 580k real-world grasping attempts, while Quillen et al. [119] train on purely simulated data. Table III shows and summarizes all RL-based works found during our review.

1) *On-Policy*: We found that out of the reviewed works employing RL, On-Policy methods were more common [112], [120], [122], [126], [129], [130], [132], [134]. In On-Policy RL, training a policy is done using experiences that are collected from the most recent policy. In the work from Kawakami et al. [130], the grasping task is divided into consecutive stages: orienting the end-effector, approaching the target, and closing the gripper. A different RL model is trained for each stage, and curriculum learning is employed that adjusts the reward function based on the success rate of each task.

Gualtieri and Platt [120] use a deep Q -network [121] with Monte Carlo updates to learn how to grasp an object and place it into a desired configuration. Working with a 24-DoF hand, Mandikal and Grauman [126] train an actor-critic model. They propose a two-step architecture: Initially, a CNN, which is trained on ContactDB [135], estimates the pixel regions that belong to a “use” affordance. An RL policy then takes this affordance mask, RGB-D image, and gripper configuration as input and outputs the wrist pose and the 24-DoF robot hand configuration. A sparse reward is awarded when the object is lifted from the ground, and a dense reward is awarded according to a distance metric from the affordance region. Chen et al. [134] used an advantage actor-critic policy gradient to train a policy that will optimize the viewpoint for grasping. They then apply the grasping algorithm developed by Pas and Platt [71] to calculate grasps for the optimized viewpoint. Wu et al. [122] use a single depth image and introduces a novel attention mechanism that learns to focus on subregions of the depth image in order to grasp better in cluttered environments. They formulate the problem using a policy gradient method based on proximal policy optimization (PPO) [123]. Wu et al. [112] extend this framework to robotic hands with arbitrary DoFs. Merzic et al. [124] proposed the use of TRPO to learn control policies that take contact feedback as input. They show that this policy significantly improved the robustness of the grasp under both object pose uncertainty and shape complexity.

2) *Off-Policy*: Off-Policy RL methods use the data collected throughout training to train a new policy. Employing human demonstrations, Song et al. [127] used Q -learning to estimate the optimal Q -function. They simulate future states by giving an action for the robot to complete, allowing the algorithm to forward simulate possible future states conditioned on the current state-action pair. Tang et al. [131] demonstrate collaborative pushing actions to facilitate grasping. Their approach uses Q -learning to learn a deterministic policy for pushing and grasping. No reward is assigned to pushing actions—the agent is only rewarded when the robot successfully grasps the object. Berscheid et al. [129] formulate the problem as a Markov decision process with a single action step. They train a fully CNN to learn a four-DoF planar grasping system. A model-based controller decides the other two DoFs by avoiding collisions and maximizing grasp quality. Wang et al. [132] trained a grasping policy from demonstrations based on the deep deterministic policy gradient algorithm [133]. The demonstrations are obtained using a motion and grasp planner, which is assumed to be an “expert” in their formulation.

D. Exemplar Methods

Exemplar methods attempt to transfer grasps from previous examples. Patten et al. [136] designed an approach to grasp novel objects by learning from experience. This is achieved using metric learning to encode objects with similar geometries nearby in feature space. Finding a successful grasp is framed as a nearest neighbor search through feature space, searching for a previously successful grasp. The approach by Mahler et al. [36]

compared a given grasp candidate against a database of successful grasps. They compare grasp candidates using three feature maps: grasp parameters, depth match gradient at local patches around the object, and similarity of the object model assessed by a deep-learning-based network. The feature maps form a prior belief distribution on the similarity to all grasps in the database. The grasp with the maximum lower confidence bound of the belief distribution is executed.

IV. SUPPORTING METHODS BASED ON DEEP LEARNING

Deep learning can be applied to methods throughout the grasping pipeline that aim to improve the success of a grasping task. This task does not have to focus solely on picking up an object. More complex manipulation tasks may require grasping that affords a particular subsequent action. For example, when handing over a full mug, the robot may need to grasp the mug handle. We found two clusters of supporting methods in the reviewed literature: shape approximation techniques and affordances based methodologies.

A. Shape Approximation

The most common form to approximate the shape of an object from partial information found in the literature is *Shape Completion*, which aims to estimate the full object model from a partial input shape (e.g. point cloud of an object from one camera view). We define *Shape approximation* more generally to include any method which approximates shape from an input. This includes the approximation of the actual shape of an object by simple(r) shapes and the fusion of multimodal data to approximate a shape.

1) *Shape Completion*: Varley et al. [137] trained a 3D CNN to employ shape completion on a single view voxel grid, outputting a voxel grid with shape completed object. Gao and Tedrake [138] instead use the method from Zhang et al. [139] for shape completion, predicting a 3D voxel grid directly from RGB-D images. Kiatos et al. [140] used a variational autoencoder [141] to predict the occluded surface points and associated normals of a partial 3D point cloud. Chavan-Dafle et al. [142] predicted the depth image that estimates the ‘back’ side of an object from a masked depth image. The front and back sides can then be stitched together quickly to form an object mesh.

The uncertainty around the output of the shape completion can be useful. Gualtieri and Platt [143] incorporate uncertainty in their shape completion network, where it represents the estimated probability that each predicted point is accurate. Another approach including uncertainty is Lundell et al. [144], which incorporates a Monte Carlo dropout procedure [145] to generate a series of shape completed objects. GraspIt! [24] is then used to plan grasps over the mean object shape. The most suitable grasp over the series of shapes is chosen as the grasp point. An example of this procedure is shown in Fig. 9. Interestingly, uncertainty is ignored in follow-on work from the same authors [51], [58], [60].

2) *Auxiliary Tasks*: In deep learning, auxiliary tasks can be added into a deep learning model, which has been shown to boost the performance of the model in some domains [146]. Jiang et al. [62] assert that 3-D reconstruction and grasping are

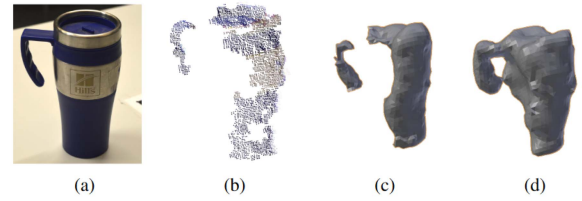


Fig. 9. Lundell et al. [144] use dropout layers to generate 20 shape completed samples, where the average of these samples is shown in (d) compared to [137](c). (©2019 IEEE).

closely related, where both rely on the knowledge of an object’s local geometries. The authors have proposed learn object reconstruction as an auxiliary task to grasping [42], [62], [88]. Jiang et al. [62] use a self-supervised approach to reconstruct an object and calculate a grasp. Yang et al. [88] simultaneously regress a grasp pose and reconstruct the point cloud of an object. The grasp pose is then refined by projecting it onto the surface of the point cloud reconstruction. Yan et al. [42] employ two networks, one for shape completion and another for grasp outcome prediction. They demonstrate a performance improvement when the grasping network uses the feature space representation produced by the shape generation network.

3) *Other*: Avigal et al. [147] do not complete explicit shape prediction; however, they use a network that takes RGB images from multiple viewpoints and generates the corresponding depth maps for the shape. The depth maps are then fed into a four-DoF grasping algorithm [148]. Ottenhaus et al. [46] used Gaussian process implicit surfaces [149] to fuse visual and tactile sensor inputs. After capturing a point cloud of the object, the robot gathers information about unseen sides of objects using tactile information. Researchers have shown that many everyday objects can be modeled as simple shapes [150], [151]. Using this observation, Torii and Hashimoto [152] approximated objects as a series of 3-D primitive shapes (hexahedron, cylinder, and sphere). They use a neural network to predict the likelihood of each primitive shape before using a precomputed database of rules to perform the grasping.

B. Affordances

This section reviews how affordances have been used in the domain of six-DoF grasping. See [153] for a review of affordances in the more general robotics domain. In addition to considering the success of grasping, these approaches have additional considerations for what kind of task it is used for. For example, when a *handover* task is to be completed for a scissors object, the robot should grasp the scissors by the blade and then pass it over to the human who can grasp it by the handle. The robot’s understanding of how an object is used in a particular task or how humans use those objects can lead to higher level reasoning about the grasping task. This lends itself toward six-DoF grasping, as different parts of the object need to be grasped in different ways, depending on the location of the affordance.

Some researchers use deep neural networks to segment objects for different affordances [126], [154], [155], [156], [157]. They then use analytical methods to find grasps within the



Fig. 10. Murali et al. [56] study task-oriented grasps on unknown objects. The top right visualization shows the grasp with the highest quality considering a given affordance. The bottom right shows all the stable grasp candidates, colored by the relevance to the affordance (green is high).

TABLE IV
NUMBER OF TIMES OBJECTS SETS ARE USED IN REVIEWED PAPERS

Object Set/Database	Number of times used	Sim/Real
YCB [159]	29	Real
3DNET [163]	15	Sim
BigBIRD [160]	12	Real
KIT [161]	12	Real
ShapeNet [162]	11	Sim
Grasp [38]	10	Sim
EGAD! [167]	2	Sim/Real
Cornell [109]	2	Real
Dex-Net [148]	2	Real
PSB [164]	1	Sim
ModelNet [165]	1	Sim
ObjectNet3D [166]	1	Sim
ContactDB [135]	1	Sim
Procedural [168]	1	Sim
Custom	11	Real/Sim

segmented object portion with an appropriate affordance. Alternatively, the approach by Murali et al. [56] estimated the quality of a sampled grasp given an affordance label. This is shown in Fig. 10, where the grasps shown in green are relevant to the given affordance. Ardón et al. [17] created a knowledge base graph representation using Markov logic networks to obtain a probability distribution of grasp affordances. In addition, the authors in [138] and [158] present an approach based on detecting a fixed number of keypoints for a category of objects.

V. DATASET DESIGN

A. Objects Sets

The objects used for training and testing grasping algorithms are crucial for the reported grasp success and allowing the community to reproduce the results. Researchers commonly use subsets of existing object sets when investigating grasping. However, there is no standard procedure for selecting this subset. This can lead to inconsistencies in objects between different works, which are using the same object set. This increases the difficulty of comparing grasping performance between works. The most commonly used object set in the reviewed works is Yale-CMU-Berkeley (YCB) [159], being used almost twice as often than the next most adopted object set (see Table IV).



Fig. 11. Commonly used object sets in robotic grasping is the YCB object set. It consists of a set of daily household objects, with a subset of the objects from the “Food” category shown here (©2015 IEEE).

YCB [159] (shown in Fig. 11), BigBIRD [160], KIT [161], and Cornell [109] consist of mostly household items such as food, toys, and tools. These object sets are appropriate for service robotics, but may not test the robustness of grasping algorithms on complex objects. ShapeNet [162], 3DNet [163], Grasp [38], PSB [164], ModelNet [165], ObjectNet3D [166], and ContactDB [135] include object model repositories, containing a large number of virtual object models, mainly used for training and testing in simulation.

EGAD! [167] and Procedural [168] consist of procedurally generated object models. EGAD! [167] proposes a set of 3-D printable objects that vary in terms of grasping difficulty [148] and object complexity [169]. Procedural [168] generate a simulated object set by attaching rectangular prisms in random orientations and locations.

Most real-world object sets do not provide a standardized method to acquire the physical objects consistently (an exception to this is YCB¹). One solution to this is 3-D printed datasets such as EGAD!² However, these objects lack semantic meaning.

B. Procedurally Generated Datasets

Even though a majority of the reviewed works use benchmark object sets, they commonly opt to create their own custom datasets with those objects. A number of works used datasets collected using real robots [63], [64], [85], [101], [129] and some combined simulation and real-world data [64], [91], [101]. However, the large majority opted to use purely simulated datasets when training their networks.

Some authors have released their datasets, with reviewed works employing public datasets such as GraspNet-1Billion [91], a hybrid six-DoF grasping dataset that captures real RGB-D camera data and combines this with simulated grasp poses, Shape Completion Grasping [137], a database of voxel grid pairs for shape completion, and ACRONYM [170], a simulation-based dataset for six-DoF grasping.

¹[Online]. Available: <https://www.ycbenchmarks.com/>

²[Online]. Available: <https://dougsml.github.io/egad/>

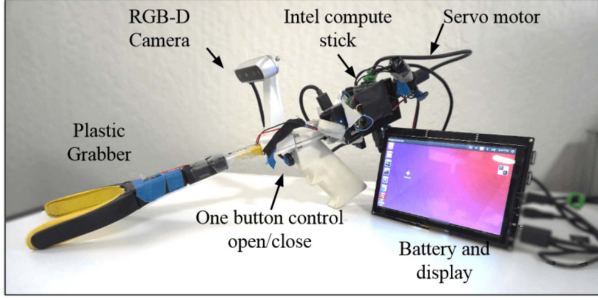


Fig. 12. Song et al. [127] design a low-cost handheld gripper to generate human annotations for grasping (©2020 IEEE).

TABLE V
NUMBER OF TIMES EACH INPUT FORMAT HAS BEEN USED

Input Format	Number of times used
Point Cloud	34
Depth Image	18
RGB-D Image	15
Voxel Grid	12
Segmentation Mask	9
Other	10

C. Expert Datasets

Researchers have proposed various datasets, which consist of expert demonstrations, either from a human or from an algorithm. Yan et al. [42] generates a dataset of around 1.6k human grasping demonstrations within virtual reality (VR) from five people. Also, in VR, Kawakami et al. [130] create a system to collect grasping demonstrations. The operator demonstrates the position of the arm using a controller with tracked position and pose. Using a handheld gripper (see Fig. 12), Song et al. [127] generate a dataset from human demonstration. The dataset contains 12 h of gripper-centric RGB-D videos, with each picking attempt separated into short clips to correspond to grasps. Similarly, Diaz Cortes et al. [171] generate 300 demonstrations by kinesthetically teaching the robot trajectories. Osorio et al. [172] create a dataset of human grasps where the human is controlling a robotic gripper with a joystick. Alternatively, the approach by Wang et al. [132] used demonstration from optimization-based motion and grasp planner [173]. Taheri et al. [174] present a dataset of whole-body grasps generated by ten subjects interacting with 51 everyday objects of varying shape and size. The object meshes are annotated with the contacts created by the human hand.

D. Data Representation

There are four major sensor inputs used for the deep learning methods: Point Cloud, Voxel Grid, RGB-D Image, and Depth Image. These four representations are all interchangeable for spatial data, assuming that intrinsic parameters for the camera are known. Table V shows the popularity of each of the input formats, and Table VI shows the popularity of different network architectures.

1) *Point Cloud*: Point clouds are the most popular data format, especially with the advancements in networks that can learn

TABLE VI
NUMBER OF TIMES EACH NETWORK BACKBONE HAS BEEN USED

Network Backbone	Number of times used
PointNet/PointNet++ [114]	21
ResNet [176]	9
LeNet [179]	3
Other	20
Custom	31

Some papers use multiple backbones, depending on their network architecture.

TABLE VII
POPULARITY OF VARIOUS TYPES OF HARDWARE SYSTEMS USED THROUGHOUT THE SURVEYED PAPERS

Category	Popularity
Robot Arm	66
Humanoid	6
Mobile Arm	3
Not Used	7

The most popular category of robotic arms is further analyzed in Table VIII.

TABLE VIII
POPULARITY OF VARIOUS ROBOTIC ARM MANUFACTURERS THROUGHOUT THE SURVEYED PAPERS

Manufacturer	Robot	Popularity	DoF
Franka Emika	Panda	15	7
Universal Robotics	UR5	9	6
Universal Robotics	UR10	4	6
Universal Robotics	UR5e	1	6
Kuka	LBR4	6	7
Kuka	IIWA LBR	3	7
Kinova	Gen3	2	7
Kinova	Mico2	2	6
Kinova	Jaco2	2	6
Rethink Robotics	Baxter	6	7
Staubli	TX60	4	6
ABB	Yumi	2	7
	Other	22	
	Custom	2	
	Not Used	7	

TABLE IX
POPULARITY OF VARIOUS GRIPPERS IN THE SURVEYED WORKS, GROUPED BY MANUFACTURER

Manufacturer	Model	Popularity	DoF
Robotiq	2F	10	1
Robotiq	3F	5	1
Franka Emika	Panda Gripper	12	1
Barret	BarretHand	7	5
Kinova	2F	3	1
Kinova	3F	1	2
Wonik Robotics	Allegro Hand	4	16
Rethink Robotics	Baxter Gripper	4	1
Shadow Robot Company	Shadow Hand	2	20
	Custom	2	
	Other	32	

directly from point clouds, such as PointNet [175]. In addition, the point cloud representation allows researchers to easily fuse data from multiple viewpoints if the relative camera poses are known.

For point-cloud-based data, researchers commonly used PointNet [175] and PointNet++ [114]. This network backbone

TABLE X
FREQUENCY OF COMMON METRICS EMPLOYED IN THE REVIEWED WORKS
AS A PERCENTAGE

Metric	Frequency Used (%)
Success Rate	86
Completion	25
Computation Time	21
Precision	12
Coverage	6
Grasp Prediction Accuracy	8

Success rate is the most commonly used metric throughout all reviewed papers.

type has been used in different methods, including direct regression [87], [89], [91], [92], [93], [99], [100], RL [132], sampling [43], [44], [49], [56], [61], and shape completion [143].

Point cloud for direct regression tends to subsample points and learn grasps for them [57], [64], [87], [89], [90], [91], [93], [97], [100], [103], [104], [105]. When a point cloud is used for sampling, one of the following procedures is usually followed. The points inside the gripper are fed to the network [44], [66], [67], the points will be transformed such that the origin aligns with the grasp frame [66], [68], or a representation of the gripper will be rendered, and all of the points will be fed into the network [43]. Minimal surveyed work has investigated the use of point clouds in RL apart from [132].

2) *Images*: The next most popular formats are depth and RGB-D images. Learning from images has been highly studied in both computer vision and robotics, allowing researchers to train preexisting architectures for robotic grasp synthesis.

For images, the most common approach is to employ ResNet [176] as the backbone [58], [60], [88], [95], [101], [102], [127], [136], [177]. In addition to ResNet, various other architectures have been used, including VGG [178] LeNet [179], DenseNet [180], and U-Net [181].

Direct regression approaches commonly use the whole image as input [84], [101], [102]. For sampling, some approaches generated a depth image slice representing points within the gripper [39], [40], while others concatenated grasp features after processing images through a CNN [35], [45], [58], [60]. RL approaches input the current camera view to generate an action [112], [120], [122], [127], [129], [131], [171].

3) *Voxel Grids*: A voxel represents a value in a regular 3-D grid. Voxel grids are analogous to images in the 3-D space, where a voxel is similar to a pixel over a 3-D grid instead of a 2-D image.

For voxel-based inputs, VoxNet [182] is most commonly used. VoxNet integrates a volumetric occupancy grid representation of the data with a 3-D CNN. Voxel grids have been used for shape completion [137], [140], sampling [52], [54], [70], and direct regression [85], [86], [94]. For sampling, voxel grids have been used similarly to point clouds, transforming the origin to align with grasp frame coordinates [52] or adding grasp configuration features after the convolutional layers [46], [54], [70]. Voxel grids have also been used to directly regress a single pose using a subset of the voxel grid corresponding to the object [85], [86] as well as directly regressing a grasp for each voxel [94].

VI. BENCHMARKING

A. Experimental Evaluation

Most approaches implemented their grasping method in the real world, either as an evaluation of the performance with respect to the metrics described in Section VI-C or presenting a demonstration that acts as a proof of concept of their method. Demonstrations focus on showing that their system works in the real-world without systematically evaluating their method. However, some approaches only consider evaluating their system using a simulator.

Evaluating in the real-world naturally carries more weight since the goal for robotic grasping is to be ultimately applied to real world. It should be noted that even though most works train their models using purely simulated data, they evaluate their approach in real-world experiments. Commonly, the grasping system was directly transferred from simulation to real world (see, e.g., [39], [40], [43], [91], [92], and [137]). Techniques such as domain adaptation [168] or domain randomization [183], [184] have also been used to transfer grasping approaches between simulation and real world. Zhu et al. [102] used contrastive learning [185] to extract invariant features when images are augmented, aiming to improve the model under image sensor noise.

B. Hardware

We found that most works that study 6-DoF grasping use a robotic arm, as shown in Table VII. However, a minority of works use other platforms, such as mobile platforms or humanoid robots. The robotic arms were most commonly table-mounted, and the robot would perform table-top grasping. Humanoid robots were commonly used when studying papers relating to affordances [84], [156], [157]. Surveyed works also completed research using mobile robots for grasping; however, none of the works made use of the extra DoFs unique to the mobile aspect of the robot.

Table IX details the frequency each robotic arm is used. Most researchers used either industrial robots or robots designed for human-robot interaction. The most common robotic platform is the Franka Emika Panda. This robot has a redundant DoF, which allows more freedom in joint angles when achieving a specific gripper pose.

The most commonly used gripper was a two-finger parallel jaw (51 times) with researchers using the gripper that came with the robotic arm. Some researchers use grippers with more than two fingers, including the Barret Hand, Allegro Hand, Shadow Hand, and Kinova 3-Finger gripper. Some papers only make use of two fingers from the Barret hand [112] or switch between two and three fingers [90]. Multifingered high-DoF hands present a different set of challenges compared to a two-finger gripper. Research on multifingered grippers focuses on how to generate a grasp pose that considers the large amount of DoF (see, e.g., [54], [66], [86], and [95]) or grasping with affordances (see, e.g., [126], [156], and [157]). Soft grippers are another potentially interesting line of work with only a few reviewed works making use of them [47], [85], [171].



Fig. 13. Examples of cluttered scenes. We differentiate Piled Clutter (left) from Structured clutter (right). Left image is from [40] (©2017 SAGE) and Right image is from [49] (©2020 IEEE).

C. Performance Metrics

A diverse set of performance metrics is used among all these works. The common definitions of performance metrics related to grasping from the reviewed work are listed below, and Table X shows the frequency each metric is used. However, the exact definition of each performance metric can vary slightly between different works:

- 1) *grasp success rate*: the percentage of successful grasps (number of successful grasps/total number of attempted grasps). The postgrasp steps prior to considering a grasp “successful” was not consistent across reviewed works;
- 2) *Completion/clearance rate*: the percentage of objects that are removed from the clutter (number of objects grasped/total number of objects in clutter);
- 3) *computation time*: time required to compute grasp hypothesis generation;
- 4) *precision*: the percentage of true positive grasp predictions (number of true positive grasp predictions/number of selected positive grasps).
- 5) *coverage*: The percentage of sampled ground truth grasps that are within a threshold distance of any of the generated grasps.
- 6) *grasp prediction accuracy*: the percentage of grasps outcomes correctly predicted (number of successful grasp predictions/number of predictions).

D. Object Configurations

Object configurations are how the objects are arranged in the scene during training or testing. We cluster object configurations into three types:

- 1) *singulated*: a single object in the scene;
- 2) *piled clutter*: objects are packed together tightly. Objects are commonly arranged as a pile, for example, Fig. 13(left);
- 3) *structured clutter*: multiple objects spread out in a scene such that they are not touching, for example, Fig. 13(right).

Piled clutter is frequently encountered in bin picking applications where a variety of items are compactly arranged in a bin. However, this survey only encompassed a limited number of papers, which address bin picking scenarios [39], [40], [59], [103], [127], [129]. In bin-picking scenarios, additional constraints are

needed for how to approach objects to avoid the bin’s walls, making it a more challenging task than table-top scenarios.

Most works surveyed in this study that focus on singulated objects tend to be grasping with a high-DOF hand [35], [37], [46], [50], [54], [57], [58], [70], [84], [86], [95], [105], [137], [144]. These papers tend to focus on solving a specific grasping task, rather than aiming for generalized grasping. Other works on singulated objects focus on data representation and the learning process [41], [92], [132], [136]. Singulated object grasping is also used in other contexts, such as affordances [17], [56], [126] or manipulation [138]. Researchers generally do not distinguish between structured or piled clutter scenes; however, we consider these distinct scenarios, which may have different solutions.

VII. CONCLUSION

This section discusses the state of the field of deep-learning-based grasp synthesis and highlights recommendations for future research directions. We note the key takeaways of each subsection indicated as *key takeaways*.

A. Deep Learning Methods

There is currently no consensus on when to use direct-regression-based methods versus sampling-based methods, as both approaches focus on similar tasks with similar success rates.

Direct-regression-based approaches have demonstrated the ability to run in real time [94], [99], but processing speeds can vary widely depending on the chosen architecture. In contrast, sampling-based methods offer the advantage of adjustable processing speed, as the number of samples or the level of optimization can be easily modified. However, based on our review of the literature, no sampling approach has been found to have real-time capabilities.

Depending on the use case of a sampling system, different sampling methods have different advantages. Euclidean sampling tends to quickly generate a wide array of grasps of varying quality. However, a large number of grasps are needed to try to find a high-quality grasp. In contrast, sampling through priors or latent space [43] tends to sample higher quality grasps; however, it can come at a cost of time to create the samples.

RL approaches for grasping typically focus not only on grasp generation but also on the trajectory used by the system. However, RL approaches typically generate only one grasp per scene, making it difficult to incorporate additional constraints during execution.

Exemplar methods, which rely on a knowledge database of objects with similar shapes, were the least commonly used approach in our literature review. This is likely because they are limited in their ability to effectively grasp objects that differ significantly from those in the database.

B. Benchmarking

Grasping in complex environments has had recent attention in adjacent fields such as agriculture [186], whereas in pure

grasping literature, the authors typically focus on grasping objects from tabletop scenes where most of the synthesized grasp poses are kinematically feasible. Studying six-DoF grasping in enclosed spaces, such as shelves [187], around obstacles [52], [59], around humans [64], near reachability limits [52], or in-the-wild (e.g., orchards [188]), would pose more constraints on the synthesis of the grasp configurations and require the grasp poses to not only be of high quality but also diverse, to increase the probability of finding a feasible trajectory to reach those grasp poses. As six-DoF grasping approaches are not often tested in such challenging environments or compared directly to four-DoF ones, the question of whether to use a six-DoF or a four-DoF grasping approach for a given application is not adequately answered by the current state of the art.

Our review also found that few research papers provide a ready-to-use implementation of their work, making it easier for other researchers to benchmark their algorithms against. On the other hand, the ones that offer an implementation are commonly used by others as part of their evaluation. For instance, work by ten Pas et al. [40] is used by many others as a benchmark, likely because it is one of the earliest works that provide an open-source implementation wrapped in a ROS package.

Key takeaways:

- 1) Six-DoF grasping should be studied in more varied environments rather than just tabletop scenarios.
- 2) Researchers should make their algorithms publicly available, ideally in a ready-to-use format (e.g., as a ROS package) to allow for informative benchmarking even on tabletop scenarios.

C. Performance Metrics

Bohg et al. [3] highlighted in 2014 that the grasping community has not yet embraced a consistent set of performance metrics. This observation is still valid today. This can be partly attributed to the large variety of objects, robots, end-effectors, and scenarios used in grasping research. Moreover, there is a divide between subsystem metrics and task-level metrics [189].

There are, however, some common performance metrics that the community has been using. The most popular metric is the grasp success rate, even though the exact definition varies in the literature. Some deem a grasp successful if the object is still held by the gripper after the robot returns to a configuration that is a certain height above the table [37], [92]. Other works impose additional constraints to the success definition, such as if the object is held for a certain amount of time [85] or checking if the object is still in the gripper after taking a sequence of actions intended to test the robustness of the grasp [51].

Although success rate is a useful metric, an underutilized type of performance metric is one that measures the time efficacy of a grasping approach. For instance, if the task is to remove objects from a container efficiently, is a slower robot with a 95% success rate preferred to a faster robot with only 75% success rate? The answer depends on the task, since faster grasping approach might be preferred for a task where dropping an object is not detrimental. One metric that could be used for this purpose is mean picks per hour (MPPH), which is defined as the average number of

successful grasps completed in an hour. Researchers should be reminded, however, that a time-based metric such as MPPH does not only measure the computational efficiency of grasp synthesis but the system as a whole and will be affected by the robot hardware, trajectory efficiency, as well as the compute resources.

The *combination* of employing both a time-based metric and success rate could provide a more comprehensive evaluation of a pure grasping system. This is due to the likely inverse nature of the two metrics, where a system with a high grasp success rate may be very computationally expensive and slow-moving. However, a very fast system may be able to perform grasps and subsequent motions quickly, however likely decreasing the grasp success, which may be undesirable when grasping more fragile objects. As such, considering both of these metrics should provide a more detailed evaluation of the grasping system.

Key takeaways:

- 1) In addition to reporting the grasp success rate, researchers should consider reporting a more strict definition of the grasp success that tests the robustness of the grasp, similar to Lundell et al. [51].
- 2) We suggest wider adoption of a time-based performance metric, such as the MPPH.

D. Object Sets and Grasping Datasets

Many papers use custom object sets, consisting of daily objects that the authors could find around the lab, making it difficult to compare different grasping approaches. Standardized object sets are very useful in enabling head-to-head comparisons of grasp synthesis algorithms. The most useful object sets for grasping research are those which cover a variety of objects and are easily accessible by the research community. Furthermore, object sets should have accurate 3-D models that enable simulation studies. The most commonly used object set currently by today's grasping community is the YCB object set [159]. However, since this may change in the future, researchers should keep an eye on which object sets the community is using.

Models for grasp synthesis are trained on datasets where each data point contains the sensor data, the grasp pose executed by the robot, along with a ground truth label depending whether the grasp was successful or not. The majority of the reviewed works were trained on simulated datasets, which offer large-scale data collection. However, there is a lack of realism due to the physics not being modeled accurately [190]. There is a lack of real-world grasping datasets, which tend to offer higher quality data and are resource intensive to create compared to simulated datasets. While we believe that a comprehensive real-world dataset could be useful for the community analogous to the commonly used Cornell Grasping dataset [109] for four-DoF top-down grasping; however, most research groups typically lack the time and resources needed to create large-scale real-world grasping datasets, with notable exceptions [117]. A compromise between real and simulated datasets is the hybrid dataset, for instance, Fang et al. [91] whom provided real-world point cloud data; however, the grasp hypotheses are evaluated analytically rather than with actual trial and error.

Key takeaways:

- 1) Grasping researchers should adopt one of the widely used object sets. YCB object set [159] is the most commonly used one today; however, in the future, new object sets might find a wider adoption.
- 2) The grasping community would benefit from the introduction of new object datasets that offer variety or specialization in different aspects such as object geometry, grasping difficulty [167], or deformation.
- 3) We propose that the authors release the code used to generate the dataset in addition to the dataset itself. This will allow other authors to make changes to the data generation procedure.

E. Trajectory Planning

Motion planning is widely utilized to find collision-free trajectories to execute a chosen grasp hypothesis. However, planning for a collision-free trajectory can be too conservative for densely cluttered scene, for example, in cabinets and shelves where it may not be possible to grasp the object of interest without nudging the neighboring objects. It has been shown that humans make deliberate contact with the environment during manipulation rather than carefully avoiding it [191]. This strategy exploits how the environment provides physical constraints on how the object and hand move and, therefore, provides a funnel for uncertainty due to noise in perception and control. This principle has also been shown to work well for robots [191], [192], [193], [194], [195], [196], [197], [198]. For example, fixturing is a widely used practice in industry for various applications such as machining, assembly, and inspection. This suggests that the objective of having collision-free trajectories to acquire a grasp must be relaxed.

Key takeaway: We encourage the study of motion planning algorithms for tight spaces where it is not possible to reach the targets without nudging other objects or where contact with the environment can be leveraged for more robust grasping.

F. Sensor Modalities

While most of the reviewed papers rely solely on vision for perceiving the world, other sensing modalities have been shown to be very helpful for both manipulation [199] and grasping [200]. Future research should explore how these techniques can be applied to improve the success of six-DoF grasping. For example, tactile sensors can be used to predict if the object will remain grasped before it is actually lifted [201], [202], [203], detect object slip [204], account for uncertainty in object pose [124], and reconstruction of object geometry [46], [205]. Force/torque sensing is another common sensor modality utilized in robotics; there are also other less common modalities including robotic skins [206] and sound [207], [208].

Key takeaway: Many of the reviewed works are vision-only sensing and considering only a single point of view. More research is needed to complement vision with other senses such as touch or hearing.

G. Grippers, Hands, and Beyond

Most of the works in this survey focus on the use of simple two- or three-fingered grippers, with few considering anthropomorphic hands. Simpler grippers greatly reduce the complexity of computing many-DoF grasp hypotheses, and, most importantly, affordable commercial grippers allow researchers to set up systems for experimentally validating their results. However, this biases the research toward problems suitable for these simpler grippers and makes research on some other problems like grasping with anthropomorphic hand, dexterous, in-hand manipulation, or soft-hand grasping fall behind. These directions may become important for enabling robots to go beyond pick and place tasks.

Key takeaway: Robots have been employed and productized in manufacturing for decades [209], [210], [211]; with further improvements in manipulation capabilities, these robots will be able to perform more complex tasks in more unstructured environments, like the assembly of unseen parts or on-site construction. Building robust and dexterous hands will result in automating many jobs currently done by humans. The opportunities are immense: new ways of designing hands, new materials, new sensors and ways of actuation, easy exchangeable fingers, and redundancy. However, it is not only about the hands—making sure that the arm can carry the hand, and that the hand is a natural extension of the arm, hand, and arms in interaction—going beyond one and compensating some DoFs in the hand by skillful dual-arm interaction are just some of the avenues to follow.

H. Grasping as Part of a Process

All day long, our hands and fingers touch, push, pull, and enclose objects. We do this with rigid, articulated, or deformable objects. We do this in the air, while objects are standing on rigid or soft surfaces or while they are moving. We do this while they are in water or covered in oil, with or without seeing—like finding a key in a pocket. Despite many decades of research and development in the area, human grasping skills are still quite superior to any of the artificial systems so far demonstrated.

In this discussion section, we have identified several areas where further contributions are needed. However, a broader view is necessary to address some if not all the manipulation scenarios mentioned above and, more importantly, consider grasping as part of an entire manipulation process. Specifically, identifying one specific six-DoF grasp pose for an object may not be sufficient to achieve success in these scenarios. One avenue may be to rethink grasp parameterizations to extend them to trajectories defined in joint and/or contact space. This could include exploiting environmental constraints [191] as well as adapting wrist trajectories based on the morphology of the hand itself [212]. New ideas and techniques for online learning and recovering from failure are required. Only new ideas and techniques in these and more areas will push advances that will make robotics systems successfully perform tasks we have been promising for some time, such as folding clothing, preparing food, and dressing humans. Pushing for and studying grasping

beyond pure grasping is one of the most important challenge we have as a community.

In summary, we have conducted a systematic review of state-of-the-art works, which use deep-learning-based toward achieving six-DoF grasping. From this review, we synthesize ten key takeaways which we believe could enable further progress in this rapidly progressing field.

APPENDIX

A. Methodology

We conducted a systematic literature review to assess the state of the art of six-DoF robotic grasping utilizing deep learning approaches. Our review, methodology adapted from [213], searched through six different scholarly libraries—IEEE, Springer, ScienceDirect, SpringerLink, arXiv, and Taylor & Francis—with the following search terms and criteria.

B. Search Terms

The following search terms were used to search through the whole paper (where the database allowed this option). The “AND” function represents a way to enforce the search results to contain all of the search terms. When possible, the search terms were combined using an “OR” function; otherwise, each term was searched separately. Furthermore, in combination with the search term, “Grasping” was searched in the metadata of the paper, if metadata search was available:

- 1) Six-DoF Grasping;
- 2) Grasping AND Point Cloud AND Deep Learning;
- 3) Shape Completion AND Six-DoF Grasping AND Learning;
- 4) Affordances AND Grasping AND Six DoFs.

C. Inclusion/Exclusion Criteria

From the papers found in the databases, only publications that met the following criteria are included in this review:

- 1) paper considered grasping from a table-top scenario;
- 2) all six DoFs were used for the grasp pose;
- 3) deep learning methods were applied in some aspect of the work;
- 4) published after January 1, 2012 (the year Alexnet [34] was published);
- 5) written in English.

All criteria were searched manually by the authors. Our search returned a total of 85 papers that matched these criteria and was, therefore, included in this review. A table of all the reviewed papers and their deep learning approaches is included in the online version of the survey.

D. Data Extraction/Analysis

Data were extracted from the included papers and verified by at least two of the authors.

ACKNOWLEDGMENT

The authors would like to thank Wesley Chan for the helpful discussion at the ideation stages of the work. The authors would also like to thank Lily Tran for helping with table creating and editing.

REFERENCES

- [1] H. Asada, “Studies on prehension and handling by robot hands with elastic fingers,” Ph.D. dissertation, Kyoto Univ., Kyoto, Japan, 1979.
- [2] A. Bicchi and V. Kumar, “Robotic grasping and contact: A review,” in *Proc. IEEE Int. Conf. Robot. Autom.*, 2000, vol. 1, pp. 348–353.
- [3] J. Bohg, A. Morales, T. Asfour, and D. Kragic, “Data-driven grasp synthesis—A survey,” *IEEE Trans. Robot.*, vol. 30, no. 2, pp. 289–309, Apr. 2014.
- [4] C. Eppner et al., “Lessons from the Amazon picking challenge: Four aspects of building robotic systems,” in *Proc. 26th Int. Joint Conf. Artif. Intell.*, 2017, pp. 4831–4835.
- [5] R. Deimel and O. Brock, “A novel type of compliant and underactuated robotic hand for dexterous grasping,” *Int. J. Robot. Res.*, vol. 35, no. 1–3, pp. 161–185, 2016.
- [6] M. G. Catalano et al., *From Soft to Adaptive Synergies: The Pisa/IIIT SoftHand*. Cham, Switzerland: Springer, 2016, pp. 101–125.
- [7] S. Levine, C. Finn, T. Darrell, and P. Abbeel, “End-to-end training of deep visuomotor policies,” *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 1334–1373, 2016.
- [8] J. Mahler et al., “Learning ambidextrous robot grasping policies,” *Sci. Robot.*, vol. 4, no. 26, 2019, Art. no. eaau4984.
- [9] O. M. Andrychowicz et al., “Learning dexterous in-hand manipulation,” *Int. J. Robot. Res.*, vol. 39, no. 1, pp. 3–20, 2020.
- [10] A. Zeng et al., “Robotic pick-and-place of novel objects in clutter with multi-affordance grasping and cross-domain image matching,” *Int. J. Robot. Res.*, vol. 41, no. 7, pp. 690–705, 2022.
- [11] A. Brohan et al., “RT-1: Robotics transformer for real-world control at scale,” 2022, *arXiv:2212.06817*.
- [12] C. Lynch et al., “Interactive language: Talking to robots in real time,” 2022, *arXiv:2210.06407*.
- [13] K. Kleeberger, R. Bormann, W. Kraus, and M. F. Huber, “A survey on learning-based robotic grasping,” *Curr. Robot. Rep.*, vol. 1, no. 4, pp. 239–249, 2020.
- [14] O. Kroemer, S. Niekum, and G. D. Konidaris, “A review of robot learning for manipulation: Challenges, representations, and algorithms,” *J. Mach. Learn. Res.*, vol. 22, pp. 1395–1476, 2021.
- [15] H. Duan, P. Wang, Y. Huang, G. Xu, W. Wei, and X. Shen, “Robotics dexterous grasping: The methods based on point cloud and deep learning,” *Front. Neurobot.*, vol. 15, 2021, Art. no. 658280.
- [16] J. J. Gibson, “The theory of affordances,” in *Perceiving, Acting, and Knowing: Toward an Ecological Psychol.*. Hillsdale, NJ, USA: Lawrence Erlbaum Associates, 1977, pp. 67–82.
- [17] P. Ardón, É. Pairet, R. Petrick, S. Ramamoorthy, and K. Lohan, “Learning grasp affordance reasoning through semantic relations,” *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 4571–4578, Oct. 2019.
- [18] A. Morales, T. Asfour, P. Azad, S. Knoop, and R. R. Dillmann, “Integrated grasp planning and visual object localization for a humanoid robot with five-fingered hands,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2006, pp. 5663–5668.
- [19] V.-D. Nguyen, “Constructing force-closure grasps in 3D,” in *Proc. IEEE Int. Conf. Robot. Autom.*, 1987, vol. 4, pp. 240–245.
- [20] K. Mirza and D. Grin, “General formulation for force distribution in power grasp,” in *Proc. IEEE Int. Conf. Robot. Autom.*, 1994, vol. 4, pp. 880–887.
- [21] I.-M. Chen and J. Burdick, “Finding antipodal point grasps on irregularly shaped objects,” *IEEE Trans. Robot. Autom.*, vol. 9, no. 4, pp. 507–512, Aug. 1993.
- [22] D. Prattichizzo and J. C. Trinkle, “Grasping,” in *Springer Handbook of Robotics*. New York, NY, USA: Springer, 2016, pp. 955–988.
- [23] R. Murray, Z. Li, and S. Sastry, *A Mathematical Introduction to Robotic Manipulation*. New York, NY, USA: Taylor & Francis, 1994.
- [24] A. Miller and P. Allen, “Grasping! A versatile simulator for robotic grasping,” *IEEE Robot. Autom. Mag.*, vol. 11, no. 4, pp. 110–122, Dec. 2004.
- [25] S. El-Khoury, A. Sahbani, and V. Perdureau, “Learning the natural grasping component of an unknown object,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2007, pp. 2957–2962.

- [26] F. Kyota, T. Watabe, S. Saito, and M. Nakajima, "Detection and evaluation of grasping positions for autonomous agents," in *Proc. Int. Conf. Cyberworlds*, 2005, pp. 453–460.
- [27] C. Michel, V. Perdureau, and M. Drouin, "An approach to extract natural grasping axes with a real 3D vision system," in *Proc. IEEE Int. Symp. Ind. Electron.*, 2006, vol. 4, pp. 3130–3135.
- [28] I. Kamon, T. Flash, and S. Edelman, "Learning to grasp using visual information," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1996, pp. 2470–2476.
- [29] Y. Jiang, S. Moseson, and A. Saxena, "Efficient grasping from RGBD images: Learning using a new rectangle representation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2011, pp. 3304–3311.
- [30] A. Saxena, J. Driemeyer, J. Kearns, C. Osundu, and A. Y. Ng, "Learning to grasp novel objects using vision," in *Experimental Robotics*. New York, NY, USA: Springer, 2008, pp. 33–42.
- [31] A. Saxena, J. Driemeyer, and A. Y. Ng, "Robotic grasping of novel objects using vision," *Int. J. Robot. Res.*, vol. 27, no. 2, pp. 157–173, 2008.
- [32] A. Saxena, L. Wong, and A. Ng, "Learning grasp strategies with partial shape information," in *Proc. Nat. Conf. Artif. Intell.*, 2008, vol. 3, pp. 1491–1494.
- [33] A. Morales, E. Chinellato, P. Sanz, A. P. del Pobil, and A. Fagg, "Learning to predict grasp reliability for a multifinger robot hand by using visual features," in *Proc. Int. Conf. AI Soft Comput.*, 2004.
- [34] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Int. Conf. Neural Inf. Process. Sys.*, 2012, vol. 25, pp. 1097–1105.
- [35] U. R. Aktaş, C. Zhao, M. Kopicki, and J. L. Wyatt, "Deep dexterous grasping of novel objects from a single view," *Int. J. Humanoid Robot.*, vol. 19, no. 2, 2022, Art. no. 2250011.
- [36] J. Mahler et al., "Dex-Net 1.0: A cloud-based network of 3D objects for robust grasp planning using a multi-armed bandit model with correlated rewards," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 1957–1964.
- [37] J. Varley, J. Weisz, J. Weiss, and P. Allen, "Generating multi-fingered robotic grasps via deep learning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2015, pp. 4415–4420.
- [38] D. Kappler, J. Bohg, and S. Schaal, "Leveraging big data for grasp planning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 4304–4311.
- [39] M. Gualtieri, A. ten Pas, K. Saenko, and R. Platt, "High precision grasp pose detection in dense clutter," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 598–605.
- [40] A. ten Pas, M. Gualtieri, K. Saenko, and R. Platt, "Grasp pose detection in point clouds," *Int. J. Robot. Res.*, vol. 36, no. 13/14, pp. 1455–1473, 2017.
- [41] Y. Zhou and K. Hauser, "6DoF grasp planning by optimizing a deep learning scoring function," in *Proc. Robot.: Sci. Syst. Workshop Revisiting Contact-Turning Probl. Solution*, 2017, vol. 2, pp. 6–11.
- [42] X. Yan et al., "Learning 6-DoF grasping interaction via deep geometry-aware 3D representations," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 3766–3773.
- [43] A. Mousavian, C. Eppner, and D. Fox, "6-DoF GraspNet: Variational grasp generation for object manipulation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 2901–2910.
- [44] H. Liang et al., "PointnetGPD: Detecting grasp configurations from point sets," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 3629–3635.
- [45] Q. Lu, K. Chenna, B. Sundaralingam, and T. Hermans, "Planning multi-fingered grasps as probabilistic inference in a learned deep network," in *Robotics Research*. New York, NY, USA: Springer, 2019, pp. 455–472.
- [46] S. Ottenhaus, D. Renninghoff, R. Grimm, F. Ferreira, and T. Asfour, "Visuo-haptic grasping of unknown objects based on Gaussian process implicit surfaces and deep learning," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2019, pp. 402–409.
- [47] J. Gonçalves and P. Lima, "Grasp planning with incomplete knowledge about the object to be grasped," in *Proc. IEEE Int. Conf. Auton. Robot Syst. Competitions*, 2019, pp. 1–6.
- [48] M. A. c. Riedlinger, M. Voelk, K. Kleeberger, M. U. Khalid, and R. Bormann, "Model-free grasp learning framework based on physical simulation," in *Proc. Int. Symp. Robot.*, 2020, pp. 1–8.
- [49] A. Murali, A. Mousavian, C. Eppner, C. Paxton, and D. Fox, "6-DoF grasping for target-driven object manipulation in clutter," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 6232–6238.
- [50] M. Van der Merwe, Q. Lu, B. Sundaralingam, M. Matak, and T. Hermans, "Learning continuous 3D reconstructions for geometrically aware grasping," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 11516–11522.
- [51] J. Lundell, F. Verdoja, and V. Kyrki, "Beyond top-grasps through scene completion," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 545–551.
- [52] X. Lou, Y. Yang, and C. Choi, "Learning to generate 6-DoF grasp poses with reachability awareness," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 1532–1538.
- [53] Y. Choi et al., "Hierarchical 6-DoF grasping with approaching direction selection," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 1553–1559.
- [54] Q. Lu, M. Van der Merwe, B. Sundaralingam, and T. Hermans, "Multi-fingered grasp planning via inference in deep neural networks: Outperforming sampling by learning differentiable models," *IEEE Robot. Autom. Mag.*, vol. 27, no. 2, pp. 55–65, Jun. 2020.
- [55] H. Schaub and A. Schöttl, "6-DoF grasp detection for unknown objects," in *Proc. Int. Conf. Adv. Comput. Inf. Technol.*, 2020, pp. 400–403.
- [56] A. Murali, W. Liu, K. Marino, S. Chernova, and A. Gupta, "Same object, different grasps: Data and semantic knowledge for task-oriented grasping," in *Proc. Conf. Robot Learn.*, 2020, pp. 1540–1557.
- [57] M. Kovic, D. Kragic, and J. Bohg, "Learning task-oriented grasping from human activity datasets," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3352–3359, Apr. 2020.
- [58] J. Lundell et al., "Multi-finGAN: Generative coarse-to-fine sampling of multi-finger grasps," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 4495–4501.
- [59] X. Lou, Y. Yang, and C. Choi, "Collision-aware target-driven object grasping in constrained environments," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6364–6370.
- [60] J. Lundell, F. Verdoja, and V. Kyrki, "DDGC: Generative deep dexterous grasping in clutter," *IEEE Robot. Autom. Lett.*, vol. 6, no. 4, pp. 6899–6906, Oct. 2021.
- [61] G. Peng, Z. Ren, H. Wang, X. Li, and M. O. Khyam, "A self-supervised learning-based 6-DoF grasp planning method for manipulator," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 3639–3648, Oct. 2022.
- [62] Z. Jiang, Y. Zhu, M. Svetlik, K. Fang, and Y. Zhu, "Synergies between affordance and geometry: 6-DoF grasp detection via implicit representations," in *Proc. Robot.: Sci. Syst. Conf.*, 2021.
- [63] H. Kasaei and M. Kasaei, "MVGrasp: Real-time multi-view 3D object grasping in highly cluttered environments," *Robot. Auton. Syst.*, vol. 160, 2023, Art. no. 104313.
- [64] X. Wang, S. Nisar, and F. Matsuno, "Robust grasp detection with incomplete point cloud and complex background," *Adv. Robot.*, vol. 35, no. 10, pp. 619–634, 2021.
- [65] M. R. Munoz, "Grasping in 6DoF: An orthographic approach to generalized grasp affordance predictions," 2021. [Online]. Available: <https://fse.studenttheses.ub.rug.nl/>
- [66] M. Corsaro, S. Tellex, and G. Konidaris, "Learning to detect multi-modal grasps for dexterous grasping in dense clutter," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 4647–4653.
- [67] Z. Ren, G. Peng, J. Yang, and H. Wang, "6-DoF grasp planning of manipulator combined with self-supervised learning," in *Proc. Chin. Control Decis. Conf.*, 2021, pp. 3026–3032.
- [68] B. Wen, W. Lian, K. Bekris, and S. Schaal, "CaTGrasp: Learning category-level task-relevant grasping in clutter from simulation," in *Proc. Int. Conf. Robot. Autom.*, 2022, pp. 6401–6408.
- [69] J. Mahler, M. Matl, X. Liu, A. Li, D. Gealy, and K. Goldberg, "DEX-Net 3.0: Computing robust vacuum suction grasp targets in point clouds using a new analytic model and deep learning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 5620–5627.
- [70] Q. Lu, M. Van der Merwe, and T. Hermans, "Multi-fingered active grasp learning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 8415–8422.
- [71] A. Pas and R. Platt, "Using geometry to detect grasp poses in 3D point clouds," *Robot. Res.*, vol. 1, pp. 307–324, 2018.
- [72] G. Smith, E. Lee, K. Goldberg, K. F. Böhlinger, and J. J. Craig, "Computing parallel-jaw grips," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1999, vol. 3, pp. 1897–1903.
- [73] S. Agrawal and N. Goyal, "Analysis of Thompson sampling for the multi-armed bandit problem," in *Proc. 25th Ann. Conf. Learn. Theory*, 2012, pp. 39.1–39.26.
- [74] N. Vahrenkamp et al., *Simox: A Robotics Toolbox for Simulation, Motion and Grasp Planning*. Berlin, Germany: Springer, 2013, pp. 585–594.
- [75] M. Ciocarlie and P. Allen, "Hand posture subspaces for dexterous robotic grasping," *Int. J. Robot. Res.*, vol. 28, pp. 851–867, 2009.
- [76] M. S. Kopicki, D. Belter, and J. L. Wyatt, "Learning better generative models for dexterous, single-view grasping of novel objects," *Int. J. Robot. Res.*, vol. 38, no. 10/11, pp. 1246–1267, 2019.
- [77] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. 2nd Int. Conf. Learn. Represent.*, 2014.

- [78] C. M. Bishop, "Mixture density networks," Aston Univ., Birmingham, U.K., Tech. Rep. NCRG/94/004, 1994.
- [79] D. Morrison, P. Corke, and J. Leitner, "Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach," in *Proc. Robot.: Sci. Syst. Conf.*, 2018.
- [80] H. Schaub, A. Schöttl, and M. Hoh, "6-DoF grasp detection for unknown objects using surface reconstruction," in *Proc. 3rd Int. Congr. Human-Comput. Interact., Optim. Robot. Appl.*, 2021, pp. 1–6.
- [81] V. Satish, J. Mahler, and K. Goldberg, "On-policy dataset synthesis for learning robot grasping policies using fully convolutional deep networks," *IEEE Robot. Autom. Lett.*, vol. 4, no. 2, pp. 1357–1364, Apr. 2019.
- [82] C. Ferrari and J. Canny, "Planning optimal grasps," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1992, pp. 2290–2295.
- [83] M. Veres, M. Moussa, and G. W. Taylor, "Modeling grasp motor imagery through deep conditional generative models," in *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 757–764, Apr. 2017.
- [84] P. Schmidt, N. Vahrenkamp, M. Wächter, and T. Asfour, "Grasping of unknown objects using deep convolutional neural networks based on depth images," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 6831–6838.
- [85] C. Choi, W. Schwarting, J. DelPreto, and D. Rus, "Learning object grasping for soft robot hands," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2370–2377, Jul. 2018.
- [86] M. Liu, Z. Pan, K. Xu, K. Ganguly, and D. Manocha, "Generating grasp poses for a high-DoF gripper using neural networks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 1518–1525.
- [87] Y. Qin, R. Chen, H. Zhu, M. Song, J. Xu, and H. Su, "S4G: A modal single-view single-shot SE (3) grasp detection in cluttered scenes," in *Proc. Conf. Robot Learn.*, 2020, pp. 53–65.
- [88] D. Yang, T. Tosun, B. Eisner, V. Isler, and D. Lee, "Robotic grasping through combined image-based grasp proposal and 3D reconstruction," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6350–6356.
- [89] K.-Y. Jeng et al., "GDN: A coarse-to-fine (C2F) representation for end-to-end 6-DoF grasp detection," in *Proc. Conf. Robot Learn.*, 2021, pp. 220–231.
- [90] C.-H. Wang and P.-C. Lin, "Q-PointNet: Intelligent stacked-objects grasping using a RGBD sensor and a dexterous hand," in *Proc. IEEE/ASME Int. Conf. Adv. Intell. Mechatronics*, 2020, pp. 601–606.
- [91] H.-S. Fang, C. Wang, M. Gou, and C. Lu, "GraspNet-1Billion: A large-scale benchmark for general object grasping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 11444–11453.
- [92] C. Wu et al., "Grasp proposal networks: An end-to-end solution for visual learning of robotic grasps," in *Proc. Conf. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 13174–13184.
- [93] P. Ni, W. Zhang, X. Zhu, and Q. Cao, "PointNet++ grasping: Learning an end-to-end spatial grasp generation algorithm from sparse point clouds," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 3619–3625.
- [94] M. Breyer, J. J. Chung, L. Ott, S. Roland, and N. Juan, "Volumetric grasping network: Real-time 6 DoF grasp detection in clutter," in *Proc. Conf. Robot Learn.*, 2020, pp. 1602–1611.
- [95] M. Liu, Z. Pan, K. Xu, K. Ganguly, and D. Manocha, "Deep differentiable grasp planner for high-DoF grippers," in *Proc. Robot.: Sci. Syst. Conf.*, 2020.
- [96] L. Shao et al., "UniGrasp: Learning a unified model to grasp with multifingered robotic hands," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 2286–2293, Apr. 2020.
- [97] P. Ni, W. Zhang, X. Zhu, and Q. Cao, "Learning an end-to-end spatial grasp generation and refinement algorithm from simulation," *Mach. Vis. Appl.*, vol. 32, no. 1, pp. 1–12, 2021.
- [98] C. Wang, H.-S. Fang, M. Gou, H. Fang, J. Gao, and C. Lu, "Graspness discovery in clutters for fast and accurate grasp detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 15964–15973.
- [99] M. Sundermeyer, A. Mousavian, R. Triebel, and D. Fox, "Contact-GraspNet: Efficient 6-DoF grasp generation in cluttered scenes," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 13438–13444.
- [100] B. Zhao, H. Zhang, X. Lan, H. Wang, Z. Tian, and N. Zheng, "REGNet: REgion-based grasp network for single-shot grasp detection in point clouds," in *IEEE Int. Conf. Robotics Automat.*, Xi'an, China, 2021, pp. 13474–13480, doi: [10.1109/ICRA48506.2021.9561920](https://doi.org/10.1109/ICRA48506.2021.9561920).
- [101] M. Gou, H.-S. Fang, Z. Zhu, S. Xu, C. Wang, and C. Lu, "RGB matters: Learning 7-DoF grasp poses on monocular RGBD images," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 13459–13466.
- [102] X. Zhu, L. Sun, Y. Fan, and M. Tomizuka, "6-DoF contrastive grasp proposal network," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6371–6377.
- [103] W. Wei et al., "GPR: Grasp pose refinement network for cluttered scenes," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 4295–4302.
- [104] Y. Li, T. Kong, R. Chu, Y. Li, P. Wang, and L. Li, "Simultaneous semantic and collision learning for 6-DoF grasp pose estimation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 3571–3578.
- [105] S. Li, Z. Li, K. Han, X. Li, Y. Xiong, and Z. Xie, "An end-to-end spatial grasp prediction model for humanoid multi-fingered hand using deep network," in *Proc. 6th Int. Conf. Control, Robot. Cybern.*, 2021, pp. 130–136.
- [106] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.
- [107] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. 28th Int. Conf. Neural Inf. Process. Syst.*, 2015, pp. 91–99.
- [108] J. Redmon and A. Angelova, "Real-time grasp detection using convolutional neural networks," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 1316–1322.
- [109] I. Lenz, H. Lee, and A. Saxena, "Deep learning for detecting robotic grasps," *Int. J. Robot. Res.*, vol. 34, no. 4/5, pp. 705–724, 2013.
- [110] F.-J. Chu, R. Xu, and P. A. Vela, "Real-world multiobject, multigrasp detection," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3355–3362, Oct. 2018.
- [111] S. Kumra and C. Kanan, "Robotic grasp detection using deep convolutional neural networks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 769–776.
- [112] B. Wu et al., "Generative attention learning: A "generAL" framework for high-performance multi-fingered grasping in clutter," *Auton. Robots*, vol. 44, pp. 971–990, 2020.
- [113] K. Sohn, H. Lee, and X. Yan, "Learning structured output representation using deep conditional generative models," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2015, pp. 3483–3491.
- [114] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet: Deep hierarchical feature learning on point sets in a metric space," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5105–5114.
- [115] Y. Li, "Deep reinforcement learning: An overview," 2017, *arXiv:1701.07274*.
- [116] M. Mohammed, L. Kwek, and S. C. Chua, "Review of deep reinforcement learning-based object grasping: Techniques, open challenges, and recommendations," *IEEE Access*, vol. 8, pp. 178450–178481, 2020.
- [117] S. Levine, P. Pastor, A. Krizhevsky, J. Ibarz, and D. Quillen, "Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection," *Int. J. Robot. Res.*, vol. 37, no. 4/5, pp. 421–436, 2018.
- [118] D. Kalashnikov et al., "QT-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation," in *Proc. Conf. Robot Learn.*, 2018.
- [119] D. Quillen, E. Jang, O. Nachum, C. Finn, J. Ibarz, and S. Levine, "Deep reinforcement learning for vision-based robotic grasping: A simulated comparative evaluation of off-policy methods," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 6284–6291.
- [120] M. Gualtieri and R. Platt, "Learning 6-DoF grasping and pick-place using attention focus," in *Proc. Conf. Robot Learn.*, 2018, pp. 477–486.
- [121] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, pp. 529–533, 2015.
- [122] B. Wu, I. Akinola, and P. K. Allen, "Pixel-attentive policy gradient for multi-fingered grasping in cluttered scenes," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 1789–1796.
- [123] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [124] H. Merzic, M. Bogdanovic, D. Kappler, L. Righetti, and J. Bohg, "Leveraging contact forces for learning to grasp," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 3615–3621.
- [125] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, "Trust region policy optimization," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, pp. 1889–1897.
- [126] P. Mandikal and K. Grauman, "Learning dexterous grasping with object-centric visual affordances," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6169–6176.
- [127] S. Song, A. Zeng, J. Lee, and T. Funkhouser, "Grasping in the wild: Learning 6DoF closed-loop grasping from low-cost demonstrations," *IEEE Robot. Autom. Lett.*, vol. 5, no. 3, pp. 4978–4985, Jul. 2020.
- [128] C. J. Watkins and P. Dayan, "Q-learning," *Mach. Learn.*, vol. 8, no. 3/4, pp. 279–292, 1992.
- [129] L. Berscheid, C. Friedrich, and T. Kröger, "Robot learning of 6 DoF grasping using model-based adaptive primitives," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 4474–4480.

- [130] D. Kawakami, R. Ishikawa, M. Roxas, Y. Sato, and T. Oishi, "Learning 6DoF grasping using reward-consistent demonstration," 2021, *arXiv:2103.12321*.
- [131] B. Tang, M. Corsaro, G. Konidakis, S. Nikolaidis, and S. Tellex, "Learning collaborative pushing and grasping policies in dense clutter," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6177–6184.
- [132] L. Wang, Y. Xiang, W. Yang, A. Mousavian, and D. Fox, "Goal-auxiliary actor-critic for 6D robotic grasping with point clouds," in *Proc. 5th Annu. Conf. Robot Learn.*, 2021, pp. 70–80.
- [133] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
- [134] X. Chen et al., "Transferable active grasping and real embodied dataset," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 3611–3618.
- [135] S. Brahmabhatt, C. Ham, C. C. Kemp, and J. Hays, "ContactDB: Analyzing and predicting grasp contact via thermal imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8709–8719.
- [136] T. Patten, K. Park, and M. Vincze, "DGCM-Net: Dense geometrical correspondence matching network for incremental experience-based robotic grasping," *Front. Robot. AI*, vol. 7, 2020, Art. no. 120.
- [137] J. Varley, C. DeChant, A. Richardson, J. Ruales, and P. Allen, "Shape completion enabled robotic grasping," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 2442–2447.
- [138] W. Gao and R. Tedrake, "KPAM-SC: Generalizable manipulation planning using keypoint affordance and shape completion," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6527–6533.
- [139] X. Zhang, Z. Zhang, C. Zhang, J. Tenenbaum, B. Freeman, and J. Wu, "Learning to reconstruct shapes from unseen classes," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 2263–2274.
- [140] M. Kiatos, S. Malassiotis, and I. Sarantopoulos, "A geometric approach for grasping unknown objects with multifingered hands," *IEEE Trans. Robot.*, vol. 37, no. 3, pp. 735–746, Jun. 2020.
- [141] D. P. Kingma and M. Welling, "An introduction to variational autoencoders," *Found. Trends Mach. Learn.*, vol. 12, no. 4, pp. 307–392, 2019.
- [142] N. Chavan-Dafle, S. Popovych, S. Agrawal, D. D. Lee, and V. Isler, "Simultaneous object reconstruction and grasp prediction using a camera-centric object shell representation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2022, pp. 1396–1403.
- [143] M. Gualtieri and R. Platt, "Robotic pick-and-place with uncertain object instance segmentation and shape completion," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1753–1760, Apr. 2021.
- [144] J. Lundell, F. Verdoja, and V. Kyriki, "Robust grasp planning over uncertain shape completions," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 1526–1532.
- [145] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1050–1059.
- [146] P. Vafaeikia, K. Namdar, and F. Khalvati, "A brief review of deep multi-task learning and auxiliary task learning," 2020, *arXiv:2007.01126*.
- [147] Y. Avigal, S. Paradis, and H. Zhang, "6-DoF grasp planning using fast 3D reconstruction and grasp quality CNN," 2020, *arXiv:2009.08618*.
- [148] J. Mahler et al., "Dex-Net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics," in *Proc. Robot.: Sci. Syst. Conf.*, 2017.
- [149] M. Björkman, Y. Bekiroglu, V. Hogman, and D. Kragic, "Enhancing visual perception of shape through tactile glances," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2013, pp. 3180–3186.
- [150] Y. Shiraki et al., "Modeling of everyday objects for semantic grasp," in *Proc. IEEE Int. Symp. Robot Hum. Interact. Commun.*, 2014, pp. 750–755.
- [151] A. Miller, S. Knoop, H. Christensen, and P. Allen, "Automatic grasp planning using shape primitives," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2003, pp. 1824–1829.
- [152] T. Torii and M. Hashimoto, "Model-less estimation method for robot grasping parameters using 3D shape primitive approximation," in *Proc. IEEE Int. Conf. Autom. Sci. Eng.*, 2018, pp. 580–585.
- [153] P. Ardón, É. Pairet, K. S. Lohan, S. Ramamoorthy, and R. Petrick, "Affordances in robotic tasks—A survey," 2020, *arXiv:2004.07400*.
- [154] Y. Li, L. Schomaker, and S. H. Kasaei, "Learning to grasp 3D objects using deep residual U-nets," in *Proc. IEEE Int. Conf. Robot Hum. Interact. Commun.*, 2020, pp. 781–787.
- [155] T.-T. Do, A. Nguyen, and I. Reid, "Affordancenet: An end-to-end deep learning approach for object affordance detection," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 5882–5889.
- [156] A. Nguyen, D. Kanoulas, D. G. Caldwell, and N. G. Tsagarakis, "Detecting object affordances with convolutional neural networks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Oct. 2016, pp. 2765–2770.
- [157] A. Nguyen, D. Kanoulas, D. G. Caldwell, and N. G. Tsagarakis, "Object-based affordances detection with convolutional neural networks and dense conditional random fields," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 5908–5915.
- [158] B. Manuelli, W. Gao, P. Florence, and R. Tedrake, "KPAM: Keypoint affordances for category-level robotic manipulation," in *Robotics Research*, T. Asfour, E. Yoshida, J. Park, H. Christensen, and O. Khatib, Eds. Cham, Switzerland: Springer, 2022, pp. 132–157.
- [159] B. Calli, A. Singh, A. Walsman, S. Srinivasa, P. Abbeel, and A. M. Dollar, "The YCB object and model set: Towards common benchmarks for manipulation research," in *Proc. Int. Conf. Adv. Robot.*, 2015, pp. 510–517.
- [160] A. Singh, J. Sha, K. S. Narayan, T. Achim, and P. Abbeel, "BigBIRD: A large-scale 3D database of object instances," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 509–516.
- [161] A. Kasper, Z. Xue, and R. Dillmann, "The kit object models database: An object model database for object recognition, localization and manipulation in service robotics," *Int. J. Robot. Res.*, vol. 31, no. 8, pp. 927–934, 2012.
- [162] A. X. Chang et al., "ShapeNet: An information-rich 3D model repository," 2015, *arXiv:1512.03012*.
- [163] W. Wohlkinger, A. Aldoma Buchaca, R. Rusu, and M. Vincze, "3DNet: Large-scale object class recognition from CAD models," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2012, pp. 5384–5391.
- [164] P. Shilane, P. Min, M. Kazhdan, and T. Funkhouser, "The Princeton shape benchmark," in *Proc. Shape Model. Appl.*, 2004, pp. 167–388.
- [165] Z. Wu et al., "3D ShapeNets: A deep representation for volumetric shapes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 1912–1920.
- [166] Y. Xiang et al., "ObjectNet3D: A large scale database for 3D object recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 160–176.
- [167] D. Morrison, P. Corke, and J. Leitner, "Egad! An evolved grasping analysis dataset for diversity and reproducibility in robotic manipulation," *IEEE Robot. Autom. Lett.*, vol. 5, no. 3, pp. 4368–4375, Jul. 2020.
- [168] K. Bousmalis et al., "Using simulation and domain adaptation to improve efficiency of deep robotic grasping," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 4243–4250.
- [169] D. Wang et al., "Adversarial grasp objects," in *Proc. IEEE 15th Int. Conf. Autom. Sci. Eng.*, 2019, pp. 241–248.
- [170] C. Eppner, A. Mousavian, and D. Fox, "Acronym: A large-scale grasp dataset based on simulation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 6222–6227.
- [171] D. S. Diaz Cortes, G. Hwang, and K.-U. Kyung, "Imitation learning based soft robotic grasping control without precise estimation of target posture," in *Proc. Int. Conf. Soft Robot.*, 2021, pp. 149–154.
- [172] V. R. Osorio et al., "37,000 human-planned robotic grasps with six degrees of freedom," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3346–3351, Apr. 2020.
- [173] L. Wang, Y. Xiang, and D. Fox, "Manipulation trajectory optimization with online grasp synthesis and selection," in *Proc. Robot.: Sci. Syst. Conf.*, 2020.
- [174] O. Taheri, N. Ghorbani, M. J. Black, and D. Tzionas, "GRAB: A dataset of whole-body human grasping of objects," in *Proc. Eur. Conf. Comput. Vis.*, 2020, 2020, pp. 581–600.
- [175] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 77–85.
- [176] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [177] M. Bajracharya et al., "A mobile manipulation system for one-shot teaching of complex tasks in homes," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 11039–11045.
- [178] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *Proc. Int. Conf. Learn. Represent.*, vol. 38, no. 10, pp. 1943–1955, 2015.
- [179] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [180] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2261–2269.
- [181] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervention*, 2015, pp. 234–241.

- [182] D. Maturana and S. Scherer, "VoxNet: A 3D convolutional neural network for real-time object recognition," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2015, pp. 922–928.
- [183] J. Tobin et al., "Domain randomization and generative models for robotic grasping," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2018, pp. 3482–3489.
- [184] S. James et al., "Sim-to-real via sim-to-sim: Data-efficient robotic grasping via randomized-to-canonical adaptation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 12619–12629.
- [185] P. Khosla et al., "Supervised contrastive learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 18661–18673.
- [186] B. Zhang, Y. Xie, J. Zhou, K. Wang, and Z. Zhang, "State-of-the-art robotic grippers, grasping and control strategies, as well as their applications in agricultural robots: A review," *Comput. Electron. Agriculture*, vol. 177, 2020, Art. no. 105694.
- [187] M. Costanzo, G. De Maria, G. Lettera, and C. Natale, "Can robots refill a supermarket shelf?: Motion planning and grasp control," *IEEE Robot. Autom. Mag.*, vol. 28, no. 2, pp. 61–73, Jun. 2021.
- [188] X. Wang, H. Kang, H. Zhou, W. Au, and C. Chen, "Geometry-aware fruit grasping estimation for robotic harvesting in apple orchards," *Comput. Electron. Agriculture*, vol. 193, 2022, Art. no. 106716.
- [189] J. Leitner et al., "The ACRV picking benchmark: A robotic shelf picking benchmark to foster reproducible research," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2017, pp. 4705–4712.
- [190] C. Rubert, D. Kappler, J. Bohg, and A. Morales, "Predicting grasp success in the real world—A study of quality metrics and human assessment," *Robot. Auton. Syst.*, vol. 121, 2019, Art. no. 103274.
- [191] C. Eppner, R. Deimel, J. Álvarez Ruiz, M. Maertens, and O. Brock, "Exploitation of environmental constraints in human and robotic grasping," *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 1021–1038, 2015.
- [192] M. Kazemi, J.-S. Valois, J. A. Bagnell, and N. Pollard, "Robust object grasping using force compliant motion primitives," in *Proc. Robot.: Sci. Syst. Conf.*, 2021.
- [193] L. Righetti et al., "An autonomous manipulation system based on force control and optimization," *Auton. Robots*, vol. 36, no. 1/2, pp. 11–30, 2014.
- [194] N. Hudson et al., "End-to-end dexterous manipulation with deliberate interactive estimation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2012, pp. 2371–2378.
- [195] M. Toussaint, N. Ratliff, J. Bohg, L. Righetti, P. Englert, and S. Schaal, "Dual execution of optimized contact interaction trajectories," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 47–54.
- [196] N. C. Daffle et al., "Extrinsic dexterity: In-hand manipulation with external forces," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 1578–1585.
- [197] N. Chavan-Daffle and A. Rodriguez, "Prehensile pushing: In-hand manipulation with push-primitives," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2015, pp. 6215–6222.
- [198] L. Shao, T. Migimatsu, and J. Bohg, "Learning to scaffold the development of robotic manipulation skills," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 5671–5677.
- [199] Z. Xia, Z. Deng, B. Fang, Y. Yang, and F. Sun, "A review on sensory perception for dexterous robotic manipulation," *Int. J. Adv. Robot. Syst.*, vol. 19, no. 2, 2022, Art. no. 17298806221095974.
- [200] Q. Bai, S. Li, J. Yang, Q. Song, Z. Li, and X. Zhang, "Object detection recognition and robot grasping based on machine learning: A survey," *IEEE Access*, vol. 8, pp. 181855–181879, 2020.
- [201] Y. Bekiroglu, J. Laaksonen, J. A. Jorgensen, V. Kyri, and D. Kragic, "Assessing grasp stability based on learning and haptic data," *IEEE Trans. Robot.*, vol. 27, no. 3, pp. 616–629, Jun. 2011.
- [202] R. Calandra et al., "More than a feeling: Learning to grasp and re-grasp using vision and touch," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3300–3307, Oct. 2018.
- [203] L. Chumbley, M. Gu, R. Newbury, J. Leitner, and A. Cosgun, "Integrating high-resolution tactile sensing into grasp stability prediction," in *Proc. Conf. Robots Vis.*, 2022, pp. 98–105.
- [204] W. Chen, H. Khamis, I. Birznieks, N. F. Lepora, and S. J. Redmond, "Tactile sensors for friction estimation and incipient slip detection—toward dexterous robotic manipulation: A review," *IEEE Sens. J.*, vol. 18, no. 22, pp. 9049–9064, Nov. 2018.
- [205] C. de Farias, N. Marturi, R. Stolkin, and Y. Bekiroglu, "Simultaneous tactile exploration and grasp refinement for unknown objects," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3349–3356, Apr. 2021.
- [206] D. Hughes, J. Lammie, and N. Correll, "A robotic skin for collision avoidance and affective touch recognition," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 1386–1393, Jul. 2018.
- [207] G. Zöllner, V. Wall, and O. Brock, "Active acoustic contact sensing for soft pneumatic actuators," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 7966–7972.
- [208] D. Gandhi, A. Gupta, and L. Pinto, "Swoosh! Rattle! Thump!—Actions that sound," in *Robotics* (ser. Robotics: Science and Systems). Cambridge, MA, USA: MIT Press, 2020.
- [209] P. Carbines, "Welding robot," U.S. Patent 335 510B1, Oct. 1999.
- [210] M. Edwards, "Robots in industry: An overview," *Appl. Ergonom.*, vol. 15, no. 1, pp. 45–53, 1984.
- [211] H. S. Cho, H. J. Warnecke, and D. G. Gweon, "Robotic assembly: A synthesizing overview," *Robotica*, vol. 5, no. 2, pp. 153–165, 1987.
- [212] K. Gilday, J. Hughes, and F. Iida, "Wrist-driven passive grasping: Interaction-based trajectory adaption with a compliant anthropomorphic hand," *Bioinspiration Biomimetics*, vol. 16, no. 2, 2021, Art. no. 026024.
- [213] M. Shishegar, D. Kerr, and J. Blake, "A systematic review of research into how robotic technology can help older people," *Smart Health*, vol. 7, pp. 1–18, 2018.



Rhys Newbury received the B.Eng. and B.Sc. degrees in computer science from Monash University, Clayton, VIC, Australia, in 2022, where he is working toward the Ph.D. degree.

His research interests include manipulation and human–robot interaction. He has a focus on systems view to problems and how to apply robots to real-world issues.



Morris Gu received the B.Eng. degree from Monash University, Clayton, VIC, Australia, in 2022, where he is working toward the Ph.D. degree.

His research interests include explainability and augmented reality within human–robot interaction.



Lachlan Chumbley is working toward the bachelor's degree with Monash University, Clayton, VIC, Australia.

His research interests include robotic grasping and manipulation, particularly though the use of tactile information.



Arsalan Mousavian received the B.Sc. degree from the Iran University of Science and Technology, Tehran, Iran, in 2010, the M.Sc. degree from the University of Tehran, Tehran, in 2013, both in artificial intelligence and robotics, and the Ph.D. degree in computer science from George Mason University, Fairfax, VA, USA, in 2018.

He is currently a Senior Research Scientist with NVIDIA Corporation, Seattle, WA, USA. His research interests include 3-D perception methods that help robots accomplish robot manipulation tasks in the real world.



Clemens Eppner received the M.Sc. in techniques for the imitation of manipulative actions by robots degree from the University of Freiburg, Freiburg im Breisgau, Germany, in 2008, and the Ph.D. degree in robot grasping by exploiting compliance and environmental constraints from the Robotics and Biology Lab, Technische Universität Berlin, Berlin, Germany, in 2018.

He is currently a Research Scientist with the Seattle Robotics Lab, NVIDIA Corporation, Seattle, WA, USA. His research interests include the problem space of grasping and manipulation, including aspects of planning, control, and perception.



Jürgen Leitner (Member, IEEE) received the B.Sc. degree in software engineering from Technische Universität Wien (Technical University of Vienna), Vienna, Austria, in 2007, the M.Sc.(Tech.) degree in space robotics from Luleå Tekniska Universitet, Luleå, Sweden, in 2009, the M.Sc. degree in space science from Aalto University, Espoo, Finland, in 2009, and Ph.D. degree from Università della Svizzera Italiana (University of Lugano), Lugano, Switzerland, in 2015.

He is the cofounder of LYRO Robotics, Clayton, VIC, Australia. He has designed, developed, and deployed robots for more than 15 years. His research interests include building intelligent robots that can safely and reliably interact with the physical world.



Jeannette Bohg (Member, IEEE) received the Ph.D. degree from the Division of Robotics, Perception and Learning, KTH Royal Institute of Technology, Stockholm, Sweden, in 2012.

She is currently an Assistant Professor of Computer Science with Stanford University, Stanford, CA, USA. She was a group leader with the Autonomous Motion Department (AMD), Max Planck Institute for Intelligent Systems, Tübingen, Germany, until September 2017. Her research interests include perception and learning for autonomous robotic manipulation and grasping.



Antonio Morales received the Ph.D. degree in computer science engineering from Universitat Jaume I, Castellón de la Plana, Spain, in 2004.

He is currently an Associate Professor with the Department of Computer Engineering and Science, Universitat Jaume I, where he is also the Director of the Degree Program on Robotics Intelligence and Vice-Dean of the School on Technology and Experimental Sciences. He is a leading Researcher with the Robotic Intelligence Laboratory, Universitat Jaume I. He is a Principal Investigator on several European and national research projects. His research interests include reactive robot grasping and manipulation.



Tamim Asfour (Senior Member, IEEE) received the diploma degree in electrical engineering and the Ph.D. degree in computer science from University of Karlsruhe, Karlsruhe, Germany, in 1994, and 2003, respectively.

He is a Professor with the Institute for Anthropomatics and Robotics, Karlsruhe Institute of Technology, Karlsruhe, Germany, where he holds the Chair of Humanoid Robotics Systems and is the Head of the High Performance Humanoid Technologies Lab. He is the developer and the leader of the development team of the ARMAR humanoid robot family. His research interests include 24/7 humanoid robotics. Specifically, he studies the mechatronics of humanoids as the synergetic integration of informatics, artificial intelligence, and mechatronics into complete humanoid robot systems, which are able to perform versatile tasks in real world.



Danica Kragic received the M.Sc. degree in mechanical engineering from the Technical University of Rijeka, Rijeka, Croatia, in 1995, and the Ph.D. degree in computer science from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2001.

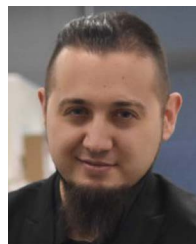
She is currently a Professor with the School of Computer Science and Communication, KTH Royal Institute of Technology.

Dr. Kragic received the 2007 IEEE Robotics and Automation Society Early Academic Career Award. She is a Member of the Swedish Royal Academy of Sciences and the Swedish Young Academy. Since 2009, she has been an AdCom Member of the IEEE Robotics and Automation Society (RAS). She has chaired the IEEE RAS Technical Committee on Computer and Robot Vision.



Dieter Fox (Fellow, IEEE) received the Ph.D. degree in computer science from the University of Bonn, Bonn, Germany, in 1998.

He is currently a Professor with the Allen School of Computer Science and Engineering, University of Washington, Seattle, WA, USA, where he heads the UW Robotics and State Estimation Lab. He is also the Senior Director of Robotics Research with NVIDIA Corporation, Seattle. He has authored or coauthored more than 200 technical papers. He is a coauthor of the textbook titled *Probabilistic Robotics* (Cambridge, MA, USA: MIT Press, 2005). His research interests include robotics and artificial intelligence, with a focus on state estimation and perception applied to problems such as mapping, object detection and tracking, manipulation, and activity recognition.



Akansel Cosgun received the Ph.D. degree in robotics from the Georgia Institute of Technology, Atlanta, GA, USA, in 2016.

He is currently a Senior Lecturer with Deakin University, Burwood, VIC, Australia. From 2018 to 2022, he was a Research Fellow with Monash University, Clayton, VIC. He conducts research in robotics, human-robot interaction, and robot learning. From 2019 to 2020, he was the Team Lead for Vision-Based Manipulation with the Australian Centre for Robotic Vision, Brisbane, QLD, Australia. He has previously worked with Honda Research, Toyota Infotechnology Center, Microsoft Research, and Savioke, a robotics start-up. His research interests include mobile robots, robotic arms, and self-driving cars with an emphasis on a system's view to problems.